Inference as Consolidation: Replay in the Silent Degrees of Freedom

Anonymous author(s)
Affiliation withheld for double-blind review

Preprint, draft of September 6, 2026

Abstract

Biological brains consolidate memories both in dedicated offline states (sleep) and, as recent causal evidence shows, through *local sleep*: brief, use-dependent off-periods of individual cortical circuits during wakefulness. In contrast, replay-based continual learning in artificial neural networks almost universally relies on offline consolidation phases or on interleaving replayed samples with the input stream. We ask whether a network trained end-to-end by local, biologically constrained learning rules can consolidate *during inference*, with no offline phase at all. We introduce (i) a *refractory rotation* rule under which units that have just fired sit out the next competition, (ii) an *isolation* scheme that confines replay updates to hidden synapses invisible to the current input under k-winner-take-all (k-WTA) dynamics — exact for silent presynaptic and suppressed units, and measured to alter the waking hidden code of only 0.2–0.4% of samples per replay step elsewhere — with per-synapse optimiser state advanced only inside the mask, and (iii) internal control signals — a unit-level homeostatic pressure that triggers replay bursts, and a relative-novelty (fast-over-slow error) gate that decides when rotation runs. The construction inverts the usual direction of non-interfering continual learning: rather than protecting past tasks while learning the present one, the hidden computation on the current input, under its suppression mask, is held invariant — exactly on the proven channels, and for all but 0.3% of waking hidden codes per update elsewhere — while past memories are consolidated through degrees of freedom the current batch exposes as unused. On class-incremental split-MNIST the resulting system reaches $\mathbf{91.6 \pm 0.3}$% (six seeds) with no offline phase, exceeding the best offline-night schedule (89.2 ± 1.7%) at 2.4× the night's replay samples (and reaching 91.1 ± 0.4% within 1.2× of that budget with eight-sample micro-batches); reversing the protected direction (guarding past memories while learning the present, as in null-space methods) costs fourteen points under identical machinery; rotation carries the gain (rotation with unmasked replay reaches 91.5 ± 0.3%; isolation adds 0.1–0.3 points at replay batches of 16 to 4, the guarantee, and at two-sample batches the absence of the collapse that unmasked replay shows in half the seeds; 2.6 points without rotation); every reported number is measured on held-out training data never used for any decision, and the headline orderings hold on a second, independent split. Rotation's i.i.d. price is governed by the substrate's activity fraction: at the 5% fraction used in the earlier phases of this work, novelty gates (masked Adam) or sleep-bout-committed gates (heavy-ball SGD) were needed to refund it, whereas at the 10% optimum the price shrinks to one to two points and both gates become neutral — increasing the activity fraction reduced the price more consistently than any tested gate. On split CIFAR-10 the ordering of methods transfers and widens, and a substrate decomposition shows that the remaining gap to backpropagation-with-replay is dense activation, not credit assignment. Finally, a drive-referenced controller for the rule's synaptic decay — one dataset-independent ratio target in place of per-dataset tuning — trades 1–3 sequential points for static gains on every CIFAR regime and lifts the system onto 100-class streams that defeat the tuned fixed decay without skips; with locally learned ResNet-style skip synapses the five-hidden-layer stack then holds the two-hidden-layer level on both axes. We provide exactness proofs for the isolation rule and release all code, checkpoints and diagnostics.

1 Problem Formulation

Setting. Let a stream of labelled examples $(x_t, y_t) \in \mathbb{R}^d \times \{1, \ldots, C\}$ be presented in T contiguous tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$, where task $\mathcal{T}_\tau$ contains only classes $\mathcal{C}_\tau \subset \{1, \ldots, C\}$, the $\mathcal{C}_\tau$ are disjoint, and no task identity is available at test time (class-incremental learning). A network f_θ with a single shared C-way readout is trained on the stream and evaluated on all C classes after the final task,

$$A_{\text{seq}}(\theta) \;=\; \Pr_{(x,y)\sim\mathcal{D}_{\text{test}}}\big[\arg\max_c f_\theta(x)_c = y\big], \qquad F \;=\; \frac{1}{T-1}\sum_{\tau=1}^{T-1}\big(a_{\tau,\tau} - a_{T,\tau}\big), \tag{1}$$

where $a_{t,\tau}$ is accuracy on task τ after learning task t and F is forgetting.

Constraints. We restrict the learner to a set $\mathcal{B}$ of cortical constraints: no weight transport (feedback synapses are learned, never copied), errors carried by a bounded burst-like channel, Dale's law, sparse population codes (k-WTA on wide layers), sparse distance-dependent connectivity, and a single forward–feedback sweep per example (no iterative relaxation). An episodic memory ("hippocampus") of at most K raw examples with reservoir writing (Vitter, 1985) is permitted; K is a budget, not a free parameter.

Objective. Rather than a single accuracy number, we optimise a four-way trade-off

$$\max_\theta \;\big(A_{\text{seq}},\, A_{\text{iid}},\, -R,\, -E\big) \quad \text{subject to} \quad \theta \in \mathcal{B}, \tag{2}$$

where A_{iid} is accuracy of the identical learner on the i.i.d. shuffle of the same data (the *static axis*: a brain-like mechanism must not damage ordinary learning), R is the replay cost in samples (synaptic work is reported separately), and E an arithmetic-energy proxy (Section 3). The central question of this paper: can the offline consolidation term be removed entirely — consolidation occurring only *during* inference on the wake stream — without losing A_{seq}, and at what cost in A_{iid} and R?

2 Related Work

Sleep-inspired consolidation in ANNs. The Bazhenov line implements sleep as a global offline phase: networks are converted to spiking dynamics and driven by noise under local Hebbian plasticity, recovering old tasks after each new one (Tadros et al., 2022), after several tasks at once (Bazhenov et al., 2026), in spiking networks with coarse alternation of training and sleep (Golden et al., 2022), and in equilibrium-propagation networks combined with small rehearsal buffers (Kubo et al., 2025). Wake–sleep frameworks with NREM and REM stages (Sorrenti et al., 2025; Robinson et al., 2022) and engineering systems such as SIESTA (Harun et al., 2023) likewise schedule consolidation offline, and SESLR (Lin et al., 2026) appends a noise-enhanced sleep phase — a spiking classifier on a frozen extractor training exclusively on its replay buffer — to correct the recency bias left by online learning. In all of these, sleep is global, offline, and clocked; none implements consolidation confined to currently unused units during ongoing inference, and the bias correction such a phase delivers is what our plastic readout receives continuously from isolated replay.

Gating, subnetwork and history-dependent competition methods. Context-dependent gating (XdG) activates a fixed random subnetwork per task but requires task identity at train and test time (Masse et al., 2018); learned gates (Tilley et al., 2023), dendritic context vectors from input prototypes (Iyer et al., 2022), task-free sparsification by feedforward and top-down errors (Lässig et al., 2023), and gradient-recovered functional masks (McKee et al., 2026) remove the explicit label but derive the gate from the *input identity*. Use-history-dependent suppression of competition, in itself, has classical precedents: refractory winner-take-all dynamics in physiological models of competitive learning (Kaski and Kohonen, 1994), explicit refractory competitive learning in which a recent winner is temporarily barred from winning (Maeda and Miyajima, 1999), and, recently, spiking continual learners whose firing thresholds rise with each unit's activation history to diversify recruitment across tasks (Shen et al., 2024). Our rotation differs in *purpose*, not primitive: recently wake-active units are benched for one competition specifically to enlarge a dynamically reallocated *replay channel* — rotation manufactures plastic degrees of freedom in which consolidation can proceed concurrently with inference, rather than allocating representational resources across tasks.

Null-space and parameter-isolation methods. A mature line of work constructs non-interfering updates by protecting *previous* tasks: projecting new-task gradients orthogonally to old-task gradient or activation subspaces (Farajtabar et al., 2020; Saha et al., 2021), training in the null space of accumulated feature covariance (Wang et al., 2021), combining k-winner sparsity and heterogeneous dropout with such projections (Abbasi et al., 2022), or freezing pruned-and-frozen subnetworks (Golkar et al., 2019; Mallya and Lazebnik, 2018). Our construction inverts the protected direction: the *ongoing wake computation* is held invariant while *old-memory replay* is written into degrees of freedom that the current batch's exact sparse support exposes as instantaneously invisible — no stored bases, SVDs, or task masks, and the "null space" changes every batch for free. McKee et al. (2026) note that optimiser state and decoupled weight decay can break structural guarantees even at zero gradient, and Bricken et al. (2023) analyse stale momentum in sparse learners; we adopt the corresponding requirement (mask-confined moments) as part of the exactness construction rather than claiming its discovery. Replay *scheduling* has likewise been studied as a learned policy (Klasson et al., 2023); our contribution there is the specific unit-level use-pressure controller and its empirical dissociation from novelty triggering.

Complementary learning systems. CLS-ER maintains fast/slow EMA copies of the weights as semantic memories (Arani et al., 2022) and DualNet trains a slow self-supervised learner beside a fast supervised one (Pham et al., 2021); both raise the value of each replayed sample but still replay at every step. Our contribution is orthogonal: we change *where* (isolated synapses), *when* (homeostatic bursts) and at what granularity (micro-batches) replay is applied.

Pattern separation. Cerebellum-, mushroom-body- and dentate-gyrus-style sparse expansions (Cayco-Gajic et al., 2017; Cayco-Gajic and Silver, 2019; Litwin-Kumar et al., 2017) inspire continual learners such as the FlyModel (Shen et al., 2021), sparse distributed memory (Bricken et al., 2023) and DG-gated mixtures (Kapoor et al., 2026). We tested this family as a front-end and report a negative result (Section 5).

Biology of local sleep. Sleep-like off-periods occur locally in awake, sleep-deprived animals (Vyazovskiy et al., 2011). Causal evidence is recent: optogenetically inducing ON/OFF alternation in one cortical hemisphere of awake mice locally discharges sleep pressure, weakens synaptic markers (GluA1, pS845) and rescues memory consolidation under sleep deprivation, whereas an equal *tonic* reduction of firing does not (Driessen et al., 2026). Our refractory rotation is the algorithmic counterpart of this alternation requirement, and our soft-rotation control (a reduction of gain rather than an off-period) the counterpart of tonic reduction.

3 Experiment Setup

Data and protocol. Sequential axis: split-MNIST (five tasks of two classes, 5 epochs per task) and split CIFAR-10 (same protocol; 32×32 luminance so that distance-dependent connectivity retains its geometry). Static axis: the identical learner and budgets on the i.i.d. shuffle. Waking batches have $n_w = 16$ samples; the episodic buffer holds $K \in \{200, 1000\}$ raw samples under reservoir writing. Forgetting F is the mean drop of each earlier task's accuracy from its own end to the end of training (per-task matrices in Appendix I); evaluation runs with the suppression mask off (k-WTA only), so test order and batch size cannot affect it. The official test sets served as the development set throughout the project; every number reported in this paper is measured on a stratified tenth of the training data held out from training and never used for any decision, with the remaining nine tenths as the stream (a second, independent tenth replicates the headline rows, Appendix I). All numbers are mean ± s.d. over three seeds unless marked otherwise; runs are deterministic at a fixed CPU thread count (the reduction order feeds k-WTA's discontinuity, so a different thread count is effectively a different seed). A third regime uses a frozen V1-like feature front-end for CIFAR-10: 256 normalised random 6×6×3 patches sampled from the training images act as fixed filters (stride 2, rectification against the per-filter mean response, quadrant average pooling; 1024-D, linear probe ≈ 49%). Nothing in the front-end is learned; the frozen representation serves as the model input. Section 7 additionally uses split CIFAR-100 (ten tasks of ten classes, same protocol and budgets, luminance and feature variants): with tenfold fewer supervision events per class it probes the learning rule far outside the regime its constants were tuned in, and its absolute numbers are floor-regime for every method tested (chance 1%, BP+ER 6–10%) — we use it for orderings and mechanisms, never as a competitive benchmark.

Architecture. d–512–256–C with k-WTA sparsity 10% on the hidden layers — the record fraction identified by the sparsity dial (Section 6.3); the project's historical 5% fraction survives in the appendices and is marked wherever used — with 30% distance-dependent connectivity on hidden synapses, Dale's law with 80% excitatory units, and the single-sweep learning rule of Section 4. The offline-night reference uses its own preferred narrower core (256–128).

Baselines. (i) Backpropagation with experience replay at equal buffer K (mandatory in every table); (ii) the offline *night*: after each waking epoch, 20 replay batches of 256 samples at 3× learning rate with no concurrent input (the strongest schedule from our earlier ablations); (iii) interleaved ER for the local learner; (iv) no buffer.

Metrics. Final accuracy, forgetting F, replay cost R (batches and samples), the isolated-channel width ρ_ℓ (fraction of replay-activated units in layer ℓ that are asleep for the current input), code overlap (mean pairwise Jaccard of task-active unit sets), participation-ratio dimensionality, and an idealised 45-nm FP32 arithmetic-energy proxy (Horowitz, 2014): 0.9 pJ per FP32 accumulate (one per synaptic event) for spiking inference vs. 4.6 pJ per multiply–accumulate for dense rate inference, after rate-to-spike conversion with thresholds calibrated on training images and T_s integration steps (Rueckauer et al., 2017). The proxy counts arithmetic only — no data movement, membrane updates or comparisons — so it is a relative measure, not a hardware figure (a fabricated neuromorphic chip reports ≈ 26 pJ per synaptic event (Merolla et al., 2014)).

4 Methodology

4.1 Cortical base learner

Layer activities are $a^0 = x$ and, for $\ell = 1, \ldots, L$,

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{3}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ is the (possibly empty) *suppression mask* of Section 4.3, and Φ_k keeps the k largest entries of each row and zeroes the rest (k-WTA); a per-unit gain exists in the implementation for homeostatic scaling but is fixed at 1 throughout. Layers $1, \ldots, L-1$ are hidden; layer L is a linear C-way readout whose pre-activation z^L is the output, scored by squared error against the one-hot target y. Errors are produced by one top-down sweep,

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big((B^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)\big]\Big), \tag{4}$$

i.e. a bounded ($|\varepsilon_i^\ell| \le \kappa \sum_j |B_{ji}^\ell|$ per component), event-gated burst signal carried by learned feedback weights $B^\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$ (no transport), in the spirit of predictive-coding and burst-dependent learners with local plasticity (Whittington and Bogacz, 2017; Payeur et al., 2021). Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment (Kolen and Pollack, 1994; Akrout et al., 2019)),

$$\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell, \qquad \Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}, \tag{5}$$

with sign-concordant feedback under Dale's law and per-synapse adaptive scaling (ϵ-floored Adam). Equations (3)–(5) were fixed by earlier ablation ladders and are not varied here, with one exception: Section 7 replaces the constant λ by an adaptive per-layer controller.

4.2 Exactly isolated replay

Let $X = \{x_b\}_{b=1}^m$ be the current waking batch, $a_{i,b}^\ell$ the activity of unit i on sample b, and $\mathcal{A}^\ell(X) = \{i : \exists b,\ a_{i,b}^\ell \neq 0\}$ the *awake set*. During the same forward step, a replay batch drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell \;=\; \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(X) \;\vee\; i \notin \mathcal{A}^\ell(X) \,\big], \tag{6}$$

i.e. a synapse may take the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. The mask applies to the *entire* replay-induced parameter change, not only to the data gradient: with heavy-ball velocity v ($\mu = 0.9$), the local update direction $G = \varepsilon^\ell (a^{\ell-1})^\top$ of (5) and the per-step decay coefficient λ, a replay step is

$$v \leftarrow M \odot (\mu v + G) + (1 - M) \odot v, \qquad W \leftarrow W + \eta\, M \odot v - \lambda\, M \odot W, \tag{7}$$

so the velocity is frozen (not zeroed) outside the mask, the decay is confined to it, and biases follow the unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell(X)]$; under Adam the first and second moments are confined the same way. As McKee et al. (2026) caution and Bricken et al. (2023) analyse for sparse learners, unmasked optimiser state would otherwise re-inject the isolated update into awake synapses, voiding the exactness guarantees below; the accuracy consequences of this choice are dissected in Section 6. The readout row and its bias remain plastic (old classes must stay calibrated); the guarantees below concern the hidden computation. Within one waking step the order is: forward pass on X under the current suppression, unmasked waking update, awake sets read from that forward pass, masked replay update on the updated weights (Algorithm 1, Appendix A); the replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and learn from it, and a burst of several micro-batches reuses its waking batch's mask.

Proposition 1 (Pre-silent updates are exactly invisible)**.** *Fix X and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta^\ell_{ij} = 0$ whenever $j \in \mathcal{A}^{\ell-1}(X)$. Then every hidden activity on every sample of X is unchanged for any magnitude of Δ; if the readout row satisfies the same condition, or is held fixed, the network output is unchanged as well.*

Proof. By induction on ℓ, for every sample b. Assume $a^{\ell-1}_{\cdot,b}$ is unchanged (true for $\ell = 1$ since $a^0_{\cdot,b} = x_b$). For any unit i, $\widehat{z}^\ell_{i,b} - z^\ell_{i,b} = \sum_j \Delta^\ell_{ij}\, a^{\ell-1}_{j,b}$, and each term vanishes: if $j \in \mathcal{A}^{\ell-1}(X)$ then $\Delta^\ell_{ij} = 0$; otherwise $a^{\ell-1}_{j,b} = 0$ for every b. Hence $z^\ell_{\cdot,b}$, and therefore $a^\ell_{\cdot,b}$ through (3), is unchanged; the induction closes at the last hidden layer, and at the output layer whenever the readout row satisfies the same condition. □

Proposition 2 (Post-asleep updates are invisible under a margin condition)**.** *Let Δ^ℓ and Δb^ℓ additionally move synapses and biases of units $i \notin \mathcal{A}^\ell(X)$ with active presynaptic partners, and write $\delta_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b} + \Delta b^\ell_i$. If for every such unit and every sample $\sigma(z^\ell_{i,b} + \delta_{i,b}) < \tau^\ell_{k,b}$, where $\tau^\ell_{k,b}$ is the k-th largest value entering Φ_k on sample b (strict, so a tie cannot promote the unit; when fewer than k entries are positive, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(\cdot) = 0$), the hidden activities on X are unchanged.*

Proof. A unit outside $\mathcal{A}^\ell(X)$ contributes $a^\ell_{i,b} = 0$ on every sample. After the update its pre-activation on sample b is $z^\ell_{i,b} + \delta_{i,b}$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a^\ell_{i,b} = 0$ still. The sample's original winners keep their pre-activations — their synapses from active presynaptic units lie outside Δ^ℓ by the mask, and their silent-presynaptic terms vanish as in Proposition 1 — so the same k units win with the same values, every other unit's input is untouched, and the layer's activities are identical. When fewer than k units are positive on sample b, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(z^\ell_{i,b} + \delta_{i,b}) = 0$: a zero-valued entry carries no activity whether or not Φ_k selects it. □

Corollary 1 (Suppressed units are exact for any magnitude)**.** *If i is suppressed by the rotation mask ($s^\ell_i = 1$ in (3)), then $a^\ell_{i,b} = 0$ for every sample and Proposition 2 holds with no margin condition.*

Remark 1 (How exact, measured)**.** *The default system keeps the readout plastic and does not enforce the margin of Proposition 2, so exactness beyond the proven channels is an empirical question, and one about the training-time computation under the suppression mask (evaluation runs with the mask off). On the record configuration, re-inferring every waking batch after each of its* 18,760 *isolated replay updates (under the suppression it was inferred with): the update altered the top hidden code of* 0.32% *of the waking samples per step (three seeds,* 0.29–0.36*), changed a waking prediction for* 0.33% *— almost all of it the plastic readout, since the same count is* 0.001% *with the readout isolated (hidden code* 0.21%*) — and violated the margin condition for* $\sim 10^{-4}$ *of the asleep units per step; the mean maximal logit change per update was* 0.02 *(*0.001 *with the readout isolated) on the one-hot scale. On raw CIFAR-10 (one seed) the hidden-code rate is* 0.46% *while the prediction rate rises to* 3.6%*, low-margin predictions flipping more easily through the plastic readout. These rates concern the hidden code of the* current *batch; what replay does to later inputs is measured by the ablation of Section 6. The isolation is thus exact for silent-presynaptic and suppressed units and near-exact elsewhere for the hidden code of the current batch; Corollary 1 is why the refractory* *rotation makes it robust rather than merely approximate. The rates are taken under the implemented order of Algorithm 1 (Appendix A), in which the awake sets come from the activities before the waking update; re-inferring the batch after that update and masking on the post-update awake sets lowers the hidden-code rate to* 0.30% *(*0.17% *with the readout isolated) and leaves the prediction rate (*0.30%*;* 0.000% *isolated), the margin-violation rate and the accuracy unchanged (*92.2 *vs.* 92.0*, one seed, development runs): the residual drift is the unenforced margin, not the ordering. (An earlier logged check that reported zero drift was taken on single batches, where the per-unit flip rate makes zero the typical outcome; the population rates above supersede it.)*

4.3 Refractory rotation and its gates

Definition 1 (Refractory rotation)**.** *After processing waking batch X_t, set $s_i^\ell(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(X_t)]$: every unit that fired sits out the next competition and is therefore asleep — and consolidable — for batch X_{t+1}.*

Rotation widens the isolated channel $\rho_\ell = |\tilde{\mathcal{A}}^\ell \setminus \mathcal{A}^\ell(X)|/|\tilde{\mathcal{A}}^\ell|$ — the fraction of the units $\tilde{\mathcal{A}}^\ell$ activated by the replay micro-batch that are asleep for the current input — from 0.2–0.3 (natural silence) to ≈ 0.53, at the price of running inference on the second-best coalition. Two internal signals control the system (Figure 1):

Homeostatic pressure (when to replay). Each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(X_t)$ with $\bar{a}_i(X_t)$ the fraction of the batch's samples on which unit i fired, and a replay *burst* of n_b consecutive micro-batches (one mask for all of them) is triggered when the mean pressure over the currently asleep units of all hidden layers exceeds θ; consolidation discharges the pressure of the units that received it, $S_i \leftarrow (1 - \delta\,\mathbf{1}[i \text{ asleep}])\,S_i$ with $\delta = 1$ throughout. This is a unit-level analogue inspired by the homeostatic Process S of sleep regulation (Borbély, 1982).

Relative novelty (when to rotate). With fast and slow EMAs of the batch-mean squared output error e_t, $\bar{e}_f(t) = (1{-}\alpha_f)\bar{e}_f + \alpha_f e_t$, $\alpha_f{=}0.1$, and $\bar{e}_s$, $\alpha_s{=}0.002$, rotation runs only while

$$\bar{e}_f(t) \;\geq\; \beta\,\bar{e}_s(t), \qquad \beta \approx 1.1\text{–}1.5, \tag{8}$$

i.e. while the stream is *new relative to the learner's own recent history* (loosely inspired by cholinergic novelty signalling (Hasselmo, 2006)). Monitoring streaming error with exponentially weighted averages is standard in concept-drift detection (Ross et al., 2012), and related loss-ratio triggers appear in concurrent continual-learning work; we claim no priority for the detector itself — only for what it gates (the rotation, hence the replay channel's width) and for the unmastered clause below. A single absolute threshold cannot serve across the regimes considered here: the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under, and we observe exactly this failure (gate open 100% of the time on static MNIST, 95% on CIFAR, for any threshold that behaves on split-MNIST). Because rotation is *beneficial* on streams the learner never masters (Section 5), the final gate adds an "unmastered" clause: with e_0 the mean error over the first 20 batches (a chance-level estimate), rotation runs while

$$\bar{e}_f(t) \;\geq\; \beta\,\bar{e}_s(t) \qquad \text{or} \qquad \bar{e}_f(t) \;\geq\; \gamma\,e_0, \qquad \gamma \approx 0.5. \tag{9}$$

The first clause opens the gate at task switches, the second keeps it open on any stream whose error remains a substantial fraction of chance — both scale-free.

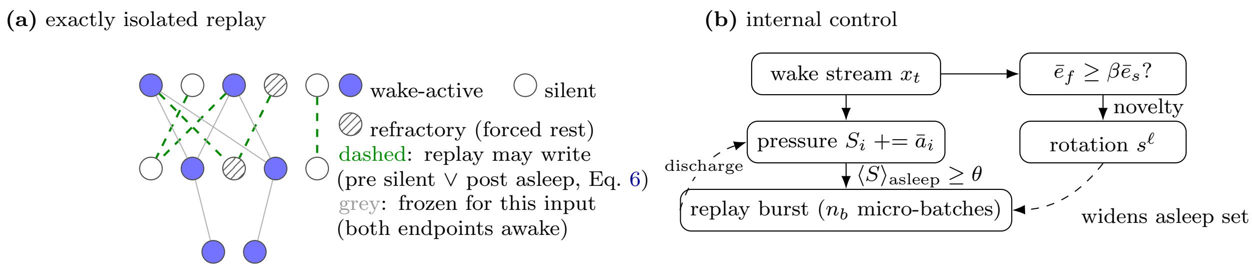

Figure 1: Mechanism. (a) During each waking batch, replay from the episodic buffer is written only into synapses whose update is invisible to the current input — exactly so for silent presynaptic and suppressed units (Propositions 1–2); refractory rotation forces recently used units into the consolidable set (Corollary 1). (b) Two internal signals: unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs. The default system runs rotation always on and replays one micro-batch at every waking batch; the pressure trigger and the novelty gate are the schedulers studied under sparse replay budgets and lower activity fractions.

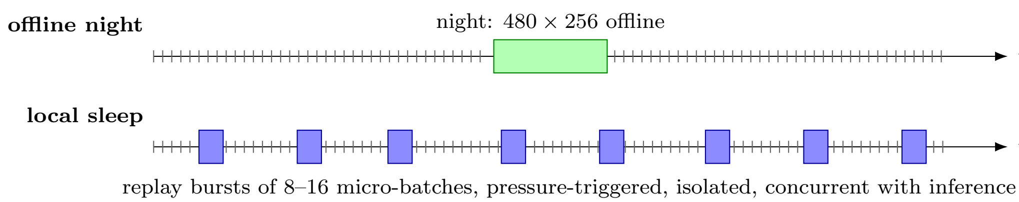

Figure 2: Consolidation schedules. Ticks are waking batches. The offline night pauses the stream; local sleep consolidates during it, in replay bursts confined to asleep units.

4.4 Cost accounting

Updates and wall-clock. Samples are not the only cost: the headline schedule makes 18,760 masked updates against the night's 480 (39×), and its training wall-clock on one CPU is 6.6 min against 0.6 for the night, 0.4 with no replay and 6.0 for unmasked interleaved replay — the per-batch replay step, not the sample count, is the online price of consolidating during the stream, in an unoptimised implementation whose mask construction is dense.

Baseline tuning and paired comparison. The offline-night reference was swept over replay batch count (20, 40 per night), replay batch size (16, 64, 256), learning-rate gain (3×), core width (256–128 and 512–256), buffer size (200, 1000, 5000) and timing (clocked after each epoch, novelty-timed, surprise-timed), and its best sequential cell is reported; BP+ER was swept over learning rate ($3{\cdot}10^{-4}$, 10^{-3}, $3{\cdot}10^{-3}$), width and buffer size; the local learner over η (Appendix F), replay batch size, cadence and the gates of Table 1. Selection everywhere was sequential accuracy at $K{=}1000$ with the static axis reported alongside, on the official test set, which served as the development set; the numbers reported are on the held-out split (Appendix I). Pairing seeds by index on that split, the local system's margin over the narrow-substrate night is +2.5 points (bootstrap 95% CI [+1.3, +4.8], three seed pairs), over the same-substrate night +2.6 ([+0.2, +5.2], six pairs; the night's seed variance makes this one fragile, and on the second split it shrinks to +0.3), over unmasked interleaved ER +6.3 ([+3.8, +9.2]), over BP+ER +2.8 ([+2.5, +3.2]), over the silent mask +3.6 ([+3.1, +4.1]) and over rotation alone +0.1 ([−0.3, +0.5]). Replay cost is measured in samples (R_s = batches × batch size) and synaptic work. The night costs $R_s = 480 \times 256 \approx 1.2 \times 10^5$ samples. The headline configuration (one 16-sample micro-batch per waking batch, 91.6 ± 0.3) costs

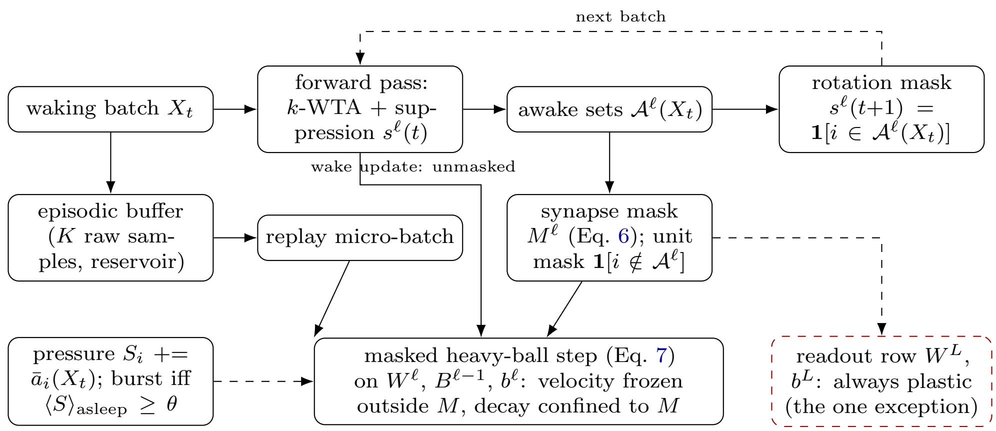

Figure 3: Training data flow. Each waking batch is inferred under the current suppression mask; its awake sets define the synapse and unit masks for the replay micro-batch applied in the same step and the rotation mask for the next batch. The masked step confines the velocity and the decay to the mask; the readout row is the one deliberately plastic exception. Unit-level pressure decides when replay bursts run.

$18{,}760 \times 16 \approx 3.0 \times 10^5$ samples ($2.4\times$ the night). Two schedules meet the night's budget within $1.2\times$: eight-sample micro-batches at every waking batch (1.5×10^5, 91.1 ± 0.4) and 16-sample micro-batches at every second one (1.5×10^5, 91.2 ± 0.2 sequential, 93.1 ± 0.2 static); pressure-triggered bursts of 16 at a third of the events reach 88.6 ± 1.6. Accuracy is bought by the frequency of small isolated updates rather than by replayed samples.

5 Results

5.1 Local sleep replaces the night

On split-MNIST at $K{=}1000$, refractory local sleep with an isolated replay micro-batch beside every waking batch reaches $\mathbf{91.6 \pm 0.3}$**%** (six seeds) with no offline phase — above the best offline night (89.2 ± 1.7% on the night's preferred narrow substrate; 89.0 ± 3.5, highly variable across seeds, on ours; paired margins $+2.5$ $[+1.3, +4.8]$ and $+2.6$ $[+0.2, +5.2]$), above unmasked interleaved ER at the same budget (87.5 ± 0.8 under Adam, 85.4 ± 3.5 under SGD) and above BP+ER (88.8 ± 0.3; $+2.8$ $[+2.5, +3.2]$). Halving the cadence with 16-sample batches costs 0.4 (91.2 ± 0.2). Adding a night on top of continuous local sleep adds 0.8 (92.4 ± 0.3 vs. 91.6 ± 0.3): once consolidation runs during wake, the offline phase is nearly redundant at this buffer size. At very small buffers ($K{=}200$) the night retains an edge only in combination with narrow substrates; the mechanism triangulation is in Appendix D.

A second split. On a second, independent held-out tenth the ordering holds (rotation 91.8 ± 0.4, BP+ER 88.5 ± 0.7, unmasked ER 80.0 ± 14.1, the narrow night 89.3 ± 0.6, the silent mask 88.6 ± 1.3, rotation alone 91.8 ± 0.6), with one exception worth stating: the same-substrate night, bimodal on the first split, reaches 91.5 ± 0.5 on the second and trails by only 0.3 there (Appendix I).

5.2 What rotation and isolation each buy

Figure 4: the full system is flat in replay batch size from 256 down to 8 samples ($91.6{\pm}0.5 \to 91.1{\pm}0.4$) and loses 1.9 at 4 ($89.7 \pm 0.7$), while at batch 16 unmasked replay without rotation sits at 85.4 ± 3.5,

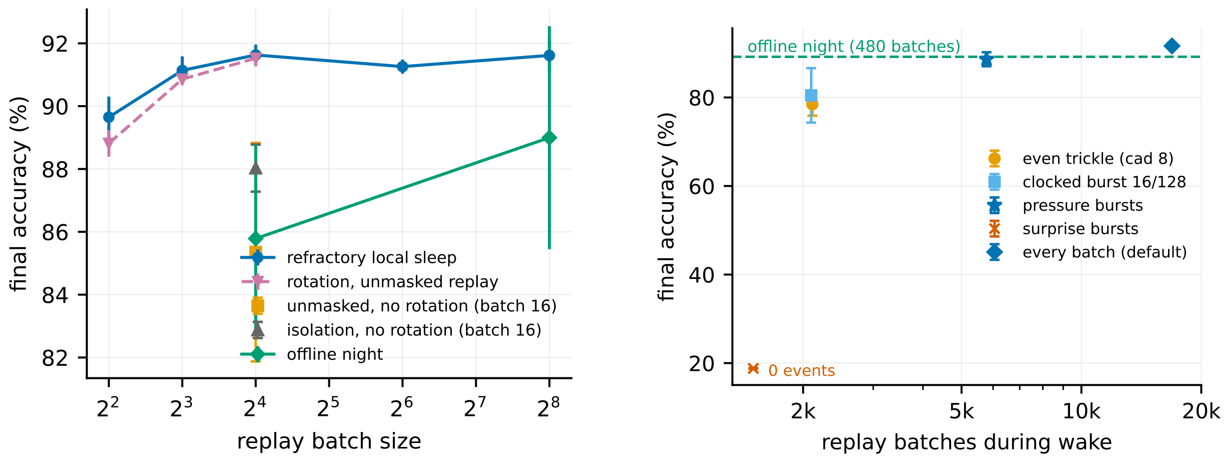

Figure 4: Left (isolation): final split-MNIST accuracy vs. replay batch size (record substrate, heavy-ball SGD); the number of updates is fixed within each series (one per waking batch; 20 per epoch for the night, whose samples therefore fall with its batch). The full system is flat down to 8 samples and loses 1.9 at 4 (12.7 at 2, Appendix H); rotation with unmasked replay (dashed) trails it by 0.1 to 0.3, within seed noise, and collapses in half the seeds at batch 2; without rotation, unmasked and isolated replay and the offline night all degrade. **Right (timing):** accuracy vs. number of waking replay batches. Trickles and clocked bursts fail alike; pressure-triggered bursts (star) keep most of the accuracy at a third of the events; surprise triggering fires no events in this regime.

isolation without rotation at 88.0 ± 0.7, and the offline night falls from 89.0 to 85.8 ± 3.0 when its batches shrink to 16. The fourth cell of the square — rotation with unmasked replay — reaches 91.5 ± 0.3 at batch 16, 90.9 ± 0.1 at 8 and 89.3 ± 0.7 at 4: the two factors are strongly sub-additive, rotation is the larger single factor (+6.1 alone, +3.6 on top of isolation), and isolation's seed-paired margin is 0.1 $[-0.3, +0.5]$ at batch 16, 0.3 $[-0.2, +0.6]$ at 8 and 0.3 $[-0.5, +1.2]$ at 4 — within seed noise at every size — with 0.4 on the static axis (94.1 vs. 93.7). At two-sample batches both systems lose heavily, but differently: isolated replay degrades gracefully (79.0 ± 3.2, six seeds within 75–84), whereas unmasked replay bifurcates — three seeds at 82–85, one at 58 and two at chance (Appendix H). Isolation's accuracy value is thus not a margin but the removal of a failure mode; what it buys at every batch size is the guarantee that, for the current input under its suppression mask, replay leaves the hidden activities unchanged — exactly on the proven channels, within the margin elsewhere; without it every micro-batch perturbs the live coalition and the outcome rests on the rotation's tolerance of that perturbation, which runs out at batch 2. The earlier "39×" cost of inference-time consolidation was an artefact of counting 256-sample batches.

5.3 The direction of protection matters, and the least optimiser state suffices

Two ablations sharpen the construction. *Mirror isolation* implements the classical protected direction with our own machinery: the *waking* update is barred from synapses whose both endpoints served the last replay batch (protect the past), while replay itself is applied unmasked. With everything else identical, mirror isolation reaches only $78.0 \pm 3.2\%$ — below even unmasked interleaved ER (87.5 ± 0.8 under Adam) and far below our direction (91.6 ± 0.3): constraining wake learning impairs the new task while the unmasked replay keeps perturbing the live computation, a double loss. This reversed-mask control thus loses fourteen points (20, with high seed variance 70.3 ± 10.1, at the historical 5% fraction, development phase); it is one instantiation of the protect-the-past direction, not a verdict on that literature. (An earlier version of this cell ran with the burst-rate baseline inadvertently frozen during the masked waking update and read 85.0 ± 2.0; the corrected cell shares its bookkeeping with every other row.) Second, replacing masked Adam with heavy-ball SGD — a single per-synapse velocity, no second-moment state — inside the same isolation gives $\mathbf{91.6 \pm 0.3}$**%** sequential and 94.1 ± 0.2% static (six seeds), a tie with masked Adam ($91.5 \pm 0.4/94.1 \pm 0.1$; unmasked SGD control 85.4 ± 3.5). Carrying a single velocity and nothing else is simpler at equal accuracy — and closer to biology; carrying *nothing* is not: with momentum zero the isolated replay micro-batches lose the velocity's noise averaging, activity collapses to chance at every learning rate unless the replay step is re-scaled by hand, and the best pure-SGD setting then trails by 3.2 sequential and 4.7 static points ($88.4 \pm 0.5/89.4 \pm 0.3$). The ablation in Section 6 finds no deficit to trace on the held-out split: confined and leaking moments tie, so the confinement of the update over time comes free. Third, *soft* rotation (halving, rather than silencing, recent winners) loses where it matters (89.9 ± 0.7 sequential, 84.3 ± 1.0 static): all-or-none OFF periods beat turned-down gain, echoing the biological finding that tonic firing-rate reduction does not discharge local sleep pressure while ON/OFF alternation does (Driessen et al., 2026). Table 1 in Section 6 collects these controls together with the remaining component knockouts; a two-sided learning-rate sweep (Appendix F) shows the comparison is a plateau rather than a knife-edge, with masked Adam already at its own best.

5.4 When replay must be rare: homeostatic bursts, not surprise

Figure 4 (right): at 12.5% of replay events, a clocked burst of 16 batches every 128 reaches 80.5 ± 6.1, no better than an even trickle at the same budget (78.5 ± 2.6). Unit-level pressure triggering keeps 88.6 ± 1.6 at a third of the events, and surprise triggering fails: at the record fraction the converged error never crosses the surprise threshold, so *zero* replay events fire (18.8, the forgetting floor). Homeostatic triggering consistently outperforms the tested surprise trigger during wake — the opposite of the night, whose best trigger in our sweeps is novelty-timed. (The same ordering held throughout the historical 5% phases, where the surprise trigger fired occasionally and still failed at every volume.)

5.5 The i.i.d. price of rotation

Rotation's i.i.d. price depends on the activity fraction. At the historical 5% it cost 1.5 points on i.i.d. MNIST (92.6 vs. 94.1 rotation-free; 1.3–3.0 across the development-phase gate grids) and, in development, *reversed* into a +2.5 to +6.5-point static gain on split CIFAR-10 (forced rotation as a regulariser in the underfit regime); at the 10% record fraction the MNIST price is 1.2–1.7 points (94.1 vs. 95.3–95.8 rotation-free) and the CIFAR static effect is inconclusive under the available seeds (28.0 ± 1.2 vs. 30.0 ± 1.9). Under masked Adam, a relative-novelty/progress gate (Eqs. (8)–(9)) removed the 5% price with one setting across all four regimes; the full Adam-era gate study — including why a single absolute threshold cannot serve across the regimes considered here and why unit-level pressure gating only trades the axes — is in Appendix C. Under heavy-ball SGD the gate had to be committed to bouts (Section 6); at the record fraction both cures become largely unnecessary — the sparsity dial (Section 6.3) shrinks the price at source.

5.6 Transfer to CIFAR-10

On split CIFAR-10 (grayscale, same protocol) the ordering transfers and widens: refractory local sleep $28.9 \pm 1.0 >$ offline night $25.7 \pm 1.2 >$ BP+ER $24.8 \pm 0.3 >$ unmasked ER $15.4 \pm 4.2 \approx$ no buffer 16.3 (the same on the second split: $29.7 > 26.5 > 25.8 > 15.5$), despite task-active code overlap of 0.9–0.98. On the V1-like feature front-end the local ordering persists (refractory 42.3 ± 0.6 $>$ offline night $34.0 \pm 0.8 \gg$ unmasked 20.5 ± 1.0) and BP+ER reaches 42.8 ± 0.7 at its best learning rate — the reverse of the raw-pixel ordering. At the 5% fraction of the earlier phases (local learner at 36.8, development phase) this was a 4.7-point regime boundary; a two-sided fairness pass plus the sparsity dial revise it to 0.5 (42.3 ± 0.6 local vs. 42.8 ± 0.7 BP+ER): most of the boundary was the local learner's optimiser and its starved activity fraction, not its learning rule. On raw pixels the local learner leads outright.

The tested decomposition is consistent with the *substrate*, not the learning rule, accounting for the residual mean gap. Transplanting the local network's ingredients into the BP net one at a time (each cell at its better of two learning rates): its width *helps* BP+ER (43.7 ± 1.0 for a dense 512–256 net), its 30% distance-dependent wiring — drawn by the same generator on the same sheets — is neutral (43.5 ± 0.2), and its k-WTA costs 3 points (40.4 ± 1.5 at the record 10%; 39.5 ± 0.6 at 5%), leaving backprop-with-replay *below* the local learner on the identical substrate (42.3 ± 0.6 vs. 40.4 ± 1.5 at 10%; three seeds each). What separates the two systems on feature inputs is dense activation, not backpropagation's credit assignment: on the substrate used here, the tested decomposition leaves no gap attributable to the learning rule. The advantage over *other local schedules* (nights, unmasked ER) remains robust across all three input regimes.

5.7 Energy

Converted to spiking inference (T_s=8; firing thresholds calibrated on training images, evaluated on the held-out tenth), the default system (always-on rotation, heavy-ball SGD, record fraction; rate accuracy 94.2% on the measured seed) runs at 92.7% and 117 nJ/sample vs. 2461 nJ for dense rate inference (21×) and 526 nJ for event-driven rate inference (4.5×), improving with the time budget (94.0% at T_s=16, 236 nJ; 94.0% at T_s=32). The rotation-free control converts to 94.2% at T_s=8 and dips to 93.9% at T_s=32; the calibration interaction we had reported under the Adam stack (the rotation-free control degrading sharply with T_s) does not reproduce at this magnitude under heavy-ball SGD on the held-out split.

5.8 Negative results

For the record, the following were falsified under matched budgets: a frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k-WTA layers: task overlap at the input drops $0.79 \rightarrow 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory *writing*; multiplicative burst coding; absolute-threshold rotation gates; pruning anchored to sparse replay bursts (Appendix E); soft (gain-reduction) rotation and mirror-direction isolation (Section 5.3); an OFF-side gate refractory — the flip-flop's inactive half in this system; strongly coupled synaptic anchors; and utility-exempt rotation as a route past the bout gate (all Section 6).

6 Ablation Study

The system under test is a conjunction of five ingredients: a biologically constrained substrate, the exact isolation mask (6), the refractory rotation, a heavy-ball optimiser, and the progress-keyed gate (9). This section removes or replaces one ingredient of the consolidation mechanism at a time (Table 1). The substrate beneath it was fixed by earlier ablation ladders; its full cost accounting — 1.2 points below a dense backpropagation MLP at 4× fewer synaptic operations per pass — is

Table 1: Component ablation of the consolidation mechanism on the record substrate (512–256/10%; backprop references dense 256–128). Sequential = final accuracy on split-MNIST; static = i.i.d. MNIST (the axis that must not degrade). Dashes: combination not applicable. Held-out split; headline rows carry six seeds, the rest three.

Variant	Sequential	i.i.d. (static)
full system: isolated replay + always-on rotation (heavy-ball SGD)	$\mathbf{91.6 \pm 0.3}$	94.1 ± 0.2
rotation gated by relative novelty, committed to bouts	91.8 ± 0.3	94.4 ± 0.3
rotation gated per batch	87.5 ± 1.6	94.7 ± 0.1
+ two-timescale anchor (λ=μ=$3{\cdot}10^{-4}$), rotation always on	91.7 ± 0.0	93.8 ± 0.3
− rotation (same mask, Eq. 6, natural silence only)	88.0 ± 0.7	95.3 ± 0.2
− isolation, − rotation (unmasked interleaved replay)	85.4 ± 3.5	95.8 ± 0.3
− isolation, + rotation (rotation alone)	91.5 ± 0.3	93.7 ± 0.3
readout-only replay (hidden layers take no replay update)	41.7 ± 10.9	82.7 ± 0.3
random rotation (matched suppression count)	91.0 ± 0.8	92.1 ± 0.2
− replay (no buffer)	19.4 ± 0.0	95.1 ± 0.0
masked Adam, moments confined to the mask	91.5 ± 0.4	94.1 ± 0.1
masked Adam, moments leaking outside the mask	91.6 ± 0.6	94.0 ± 0.3
momentum 0 (pure SGD; η and replay gain re-tuned, else chance)	88.4 ± 0.5	89.4 ± 0.3
offline night instead of concurrent replay, same substrate	89.0 ± 3.5	—
the same night on its preferred narrow substrate	89.2 ± 1.7	—
mirror direction: protect the past	78.0 ± 3.2	91.7 ± 0.6
soft rotation: gain 0.5 instead of 0	89.9 ± 0.7	84.3 ± 1.0
unmasked ER at the same budget (Adam)	87.5 ± 0.8	95.1 ± 0.2
backprop (+ER sequential; plain static), dense	88.8 ± 0.3	97.4 ± 0.1

reproduced in Appendix B. Every mechanism cell shares the record substrate (512–256 hidden, 10% k-WTA, 30% distance-dependent connectivity), the same buffer (K=1000, random writing) and the same replay budget (waking batch 16, replay batch 16 after every waking batch); the appendices retain the corresponding grids at the historical 5% fraction, where every ordering below was first established.

6.1 Reading the component table

Read top-down, the first block prices each ingredient by what its removal costs the sequential axis under heavy-ball SGD: rotation is worth +3.6 points (91.6 vs. 88.0 for the same mask on natural silence alone), rotation and isolation together +6.2 with a twelve-fold smaller seed deviation (91.6 ± 0.3 vs. 85.4 ± 3.5), isolation alone +2.6 (88.0 vs. 85.4), isolation on top of rotation +0.1 (91.5 ± 0.3 for rotation with unmasked replay; $[-0.3, +0.5]$ paired), and replay is worth everything (19.4 is the forgetting floor). Replay confined to the readout row, hidden layers frozen for replay, falls to 41.7 ± 10.9 (ten classes) and costs 11 static points: readout-only replay cannot account for the full system's accuracy, and hidden-layer replay updates are essential to it. A random suppression of matched size in place of the refractory rule recovers most of rotation's gain (91.0 ± 0.8) but trails it by 0.6 sequential and 2.0 static points with 1.4× the forgetting (F=8.7 vs. 6.4): alternating the coalition carries most of the effect, resting the recent winners the rest. The static column identifies who pays: rotation is the only statically costly component (94.1 always-on against 95.3–95.8 without it, and 93.7 with rotation and no isolation), a price of 1.2–1.7 points at the record fraction. The unmasked control is not handicapped by the replay step (3η against a waking step of $\eta/16$): sweeping the replay gain over $\{1/16, 1/4, 1, 3, 10\}$ moves masked replay $67.2 \to 86.8 \to 90.8 \to 91.6 \to 74.2$ and unmasked replay $61.6 \to 81.4 \to 85.7 \to 85.4 \to 47.3$ (sequential; three seeds, six at gain 10, where both destabilise) — both peak at the default and the rotation-free unmasked control never closes the gap; the sweep bounds that control, it does not estimate isolation's own contribution (the square above does). The gates that were built to refund a larger price at the historical 5% fraction are here *neutral*: the bout-committed gate trades +0.2 sequential for +0.3 static ($91.8 \pm 0.3/94.4 \pm 0.3$), the per-batch gate is erratic ($87.5 \pm 1.6/94.7 \pm 0.1$), and the two-timescale anchor no longer adds anything ($91.7 \pm 0.0/93.8 \pm 0.3$). Their mechanisms remain instructive — at 5% the per-batch gate's flicker destroyed heavy-ball SGD's gain, episode commitment (loosely modelled on the consolidated bouts of the sleep–wake switch (Saper et al., 2005), which motivates commitment but not the specific minimum duration) repaired it, and bout-length and OFF-refractory controls located the residual price in the rotation of settled coalitions itself — and that full study is preserved in Appendix G and the 5% grids; but at the record fraction the dial has absorbed most of what they bought, and the default system is simply always-on rotation.

The middle block tests the confinement requirement itself. Propositions 1–2 require Adam's moments to advance only inside the mask: state advancing outside re-injects the replay gradient into awake synapses through the next unmasked waking step: the present batch's invariance still holds, since the weight change itself stays masked, but the confinement of the update over time is lost. Empirically, on the held-out split, confined and leaking moments tie (91.5 ± 0.4 vs. 91.6 ± 0.6), as does the single velocity (91.6 ± 0.3): in development the leak had appeared to help by 0.8, a margin that did not survive the protocol change. The velocity is kept as the least state that stays confined to the mask at equal accuracy: it advances only inside the mask, and removing it as well (momentum zero, Table 1) costs 3.2/4.7 points once the replay step has been re-scaled to survive at all.

One budget note completes the block: half the cadence with 16-sample batches costs 0.4 (91.2 ± 0.2) — the system prefers consolidation to run continuously.

6.2 Two probes of the residual static price

At the historical 5% fraction, the OFF-refractory null located the bout gate's residual static price in the rotation of settled coalitions itself. Two mechanisms attacked that diagnosis directly (details in Appendix G; 5% numbers). *Utility-exempt rotation* (sparing the top-q long-use units from ever being benched) only traces a trade-off that the bout gate dominates: temporal commitment beats structural exemption. *Two-timescale synapses* in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016) fail to dissolve the price, but weak symmetric coupling acts as a *sequential stabiliser* at 5% (92.4 ± 0.2, variance halved, static parity); at the record fraction the anchor is neutral ($91.7 \pm 0.0/93.8 \pm 0.3$). Anchor and bout gate compose sub-additively. Figure 5 draws the operating frontier.

6.3 The sparsity dial

The substrate's original 5% k-WTA fraction was fixed in an early phase and inherited by a year of experiments; the decomposition just given made it suspect — if sparse k-WTA taxes BP by points on features, it presumably taxes our own learner too. It did, on both regimes, and the tax was bought back almost for free; this subsection documents the dial that led to the 10% record fraction used throughout the main text. On feature CIFAR-10, relaxing the fraction improves *both* axes monotonically: sequential $40.9 \to 42.9 \pm 0.6$ and static $38.3 \to 46.4 \pm 1.6$ from 5 to 25%, with forgetting slightly *reduced*; the BP control traces the same curve ($39.5 \to 40.4 \to 40.7$ at 5, 10, 15%),

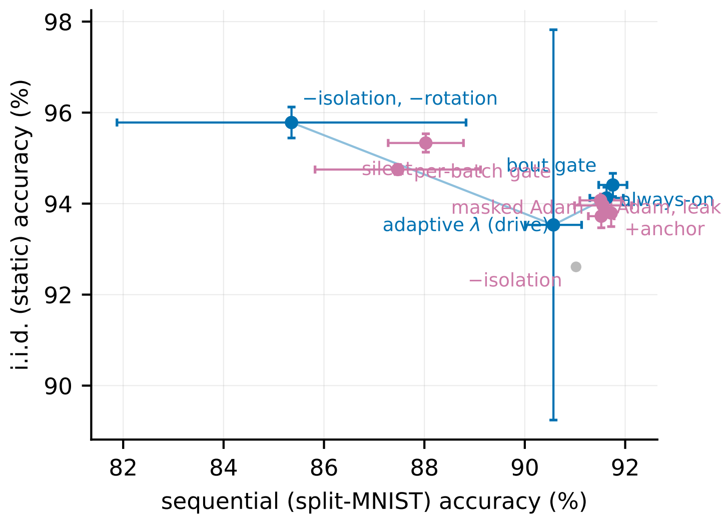

Figure 5: The operating frontier at the record fraction under heavy-ball SGD: split-MNIST sequential accuracy against i.i.d. static accuracy (headline points six seeds). Blue: non-dominated configurations; purple: dominated ones; held-out split. The adaptive-λ point (Section 7) sits on the static side of the frontier. No single configuration wins both axes; always-on rotation is the sequential default.

with the local learner 1.4–1.9 points ahead at every matched level. On MNIST the sequential axis is flat from 10%: always-on rotation reaches $\mathbf{91.6 \pm 0.3}$ sequential at 94.1 ± 0.2 static (six seeds), 15% gives $91.4 \pm 0.4/94.4 \pm 0.2$ and 25% $91.5 \pm 0.4/94.6 \pm 0.2$, against $91.0 \pm 0.5/92.6 \pm 0.0$ at the historical 5%.

Three controls sharpen the reading. First, the no-buffer ceiling itself rises with the fraction (94.1 at 5%, 95.1 at 10%), so rotation's static price *at matched sparsity* shrinks from 1.5 to 1.0 points — and the gate's raison d'être shrinks with it: at 10% always-on rotation no longer trails the bout-gated system. Second, in development, rotation-free silent isolation gained nothing from the dial (89.0 → 89.3 → 88.7, official test set): the dial helps specifically the *rotating* family, which at 5% was starved of representational room. Third, asymmetric schedules (a mild input-facing layer above a sparse deep layer) failed in development — sequential parity with uniform relaxation but a 3–4-point static deficit: the tax is levied at every layer. The dial is free elsewhere: synaptic operations rise only 3–5%. Following this finding the record substrate moved to 10%: Table 1 and every main-text number now use it, with the complete mechanism grid re-established there (orderings preserved throughout), while the appendices retain the historical 5% grids on which each mechanism was first isolated. The dial, not the gate, is the cheapest cure for rotation's static price (Figure 5).

7 An adaptive decay controller

The last tuned constant. Every result above runs the rule of (5) with a decay λ tuned once per dataset (10^{-3} on the image benchmarks). Split CIFAR-100 — ten tasks of ten classes — exposes what that constant hides. With one hundred classes, each class receives a tenth of the per-epoch supervision events, while the decay acts on every synapse at every step, undiminished; at the MNIST-tuned value the hidden-to-hidden weights follow the pure-decay trajectory $(1-\lambda)^t$ *exactly* (the thinned error signal loses the race outright), deep activity collapses by 5×, the readout gradient goes to zero, and the network lands at chance — 2.4% i.i.d. against 20–23% for a ten-class subset

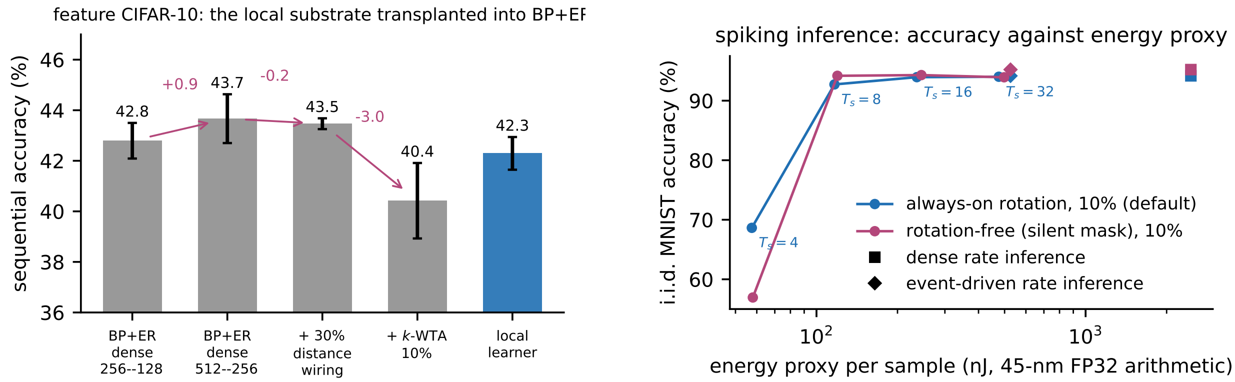

Figure 6: Left: the substrate decomposition on feature CIFAR-10 — transplanting the local network's ingredients into BP+ER one at a time (each cell at its better of two learning rates; six seeds except the local learner, three). **Right:** spiking inference, i.i.d. MNIST accuracy against the idealised 45-nm FP32 arithmetic-energy proxy for $T_s \in \{4, 8, 16, 32\}$, with the dense (square) and event-driven (diamond) rate references; held-out split; the default is monotone in T_s, the rotation-free control dips 0.4 at T_s=32.

of the same images on the same substrate (development-phase diagnosis). The class count of the supervision signal is a hidden variable that every fixed λ is silently tuned against. A manual rescue (λ=10^{-4}, 3× targets) revives activity but survives the full protocol only at 1.0–1.2%.

The controller. The project's design principle — control signals must be self-referenced, never absolute (Section 4.3) — applies at the synapse. We replace the constant by a per-layer ratio controller,

$$\lambda_\ell \;=\; \min\Bigl(\lambda_{\max},\; \rho\,\frac{\mathrm{EMA}\bigl[\,u^\ell\,\bigr]}{\overline{|W^\ell|}+\epsilon}\Bigr), \qquad \rho = 0.25,\;\; \lambda_{\max} = 0.02, \tag{10}$$

where u^ℓ is the mean magnitude of the data-driven, post-optimiser, *pre-decay* waking increment applied to layer ℓ (so the decay never feeds its own drive), the EMA (coefficient 0.02) runs over waking, unmasked steps only, $\epsilon = 10^{-12}$ floors the denominator, and W^ℓ and $B^{\ell-1}$ share λ_ℓ, so Kolen–Pollack alignment is untouched. The decay's mean magnitude per step, $\lambda_\ell\overline{|W^\ell|}$, is held at the fraction ρ of the mean data-driven update magnitude: layers and regimes with rich supervision keep a strong regulariser, starved ones are automatically spared. ρ is the only control target; $\lambda_{\max}$, the EMA coefficient and ϵ are global safeguards never tuned per dataset. Isolation and rotation decide *where* and *when* to consolidate; the controller decides *how strongly* to update as the scale changes, by the principle behind every control signal in this paper — novelty as a fast-over-slow error ratio, homeostasis as a unit's own use, decay as a drive-over-weight ratio: all referenced to the learner's own current state, none absolute. A gradient-referenced variant ($\rho\,\eta\,\mathrm{EMA}[\,\overline{|g^\ell|}\,]/\overline{|W^\ell|}$) behaved equivalently in development (official test set, not re-run under the held-out protocol); the drive-referenced form is canonical because it is optimiser-agnostic.

One ratio target, every dataset. Table 2 gives the cross-dataset verdict. On the sequential axis the controller concedes 1–3 points to the per-dataset tuned value; on the static axis it gains 1.2–5.3 points on CIFAR-10, feature CIFAR-10 and both CIFAR-100 variants, while on split-MNIST it gains 1.1 on the second held-out split (95.2 ± 0.4 vs. 94.1 ± 0.6) but one of six seeds collapsed on the first (93.5 ± 4.3 vs. 94.1 ± 0.2), so we claim no static gain there. On CIFAR-100 the controller lifts the system 2.7–4.7× over the manual rescue with no dataset-specific decay coefficient (floor-regime

Table 2: One dataset-independent ratio target replaces per-dataset decay tuning. Sequential / static accuracy (%) under the tuned fixed decay (10^{-3} on the image benchmarks; 10^{-4} with 3× targets as the CIFAR-100 rescue) against the drive-referenced controller of (10) at the same $\rho = 0.25$ everywhere; held-out split, three seeds (six on split-MNIST); the full system (refractory rotation, isolated replay, K=1000) in every cell.

	fixed λ (tuned)		controller (drive)	
	seq.	static	seq.	static
split-MNIST	91.6 ± 0.3	94.1 ± 0.2	90.6 ± 0.6	93.5 ± 4.3
split CIFAR-10	28.9 ± 1.0	28.0 ± 1.2	26.2 ± 1.1	30.1 ± 0.2
feature CIFAR-10	42.3 ± 0.6	42.4 ± 1.1	39.1 ± 0.4	43.6 ± 0.6
split CIFAR-100	1.2 ± 0.1	1.0 ± 0.1	3.2 ± 0.8	4.1 ± 0.6
+ features	1.1 ± 0.3	1.1 ± 0.2	5.2 ± 1.6	6.4 ± 0.6

numbers, reported for ordering only: the strong BP+ER reference reaches 6.8 raw and 10.1 with features). The realised λ_ℓ (logged throughout) self-differentiates exactly as the tuning history would predict — $\sim 10^{-5}$ on MNIST sequential streams, $\sim 10^{-4}$ on static ones, up to 3×10^{-3} on raw CIFAR, layer-graded within each net — so the per-dataset λ ladder was a signal-to-decay ratio in disguise. On the operating frontier (Figure 5) the controller point sits on the static side; the tuned fixed decay remains the sequential-axis record holder.

Depth is the same boundary, and two mechanisms cross it. The class count thins the error signal per class; depth attenuates it per layer — the same starvation on a different axis. A depth ladder (three, four and five hidden layers, 512–256–128 . . ., everything else the record configuration) makes the parallel exact: under the tuned fixed decay without skips the static axis dies at four hidden layers ($10.1 \pm 0.2\%$, chance) and the sequential axis collapses at five (26.3 ± 20.2), while under the controller every cell stays alive and degrades gracefully — $90.7 \to 88.9 \to 88.1$ sequential, $95.3 \to 94.8 \to 89.6$ static — with the realised λ_ℓ dropping to 3×10^{-6} in the starved middle layers. The residual toll is then architectural, and a biologically unexceptional bypass removes it. ResNet-style skip synapses — S^ℓ carrying $a^{\ell-2}$ into each deep hidden layer, learned by the same local product with a Kolen–Pollack feedback twin and the shared per-layer decay, and replay-masked under the same pre-or-post-asleep rule (Appendix A) — restore the thick error path, and the five-hidden-layer system reaches 91.0 ± 0.3 sequential and 95.8 ± 0.3 static (six seeds) — against the two-layer default's 91.6/94.1 and the two-layer controller's 90.6/93.5: 0.6 sequential points below the default and 1.7 above it on the static axis, with 95.8 ± 0.1 at depth four. The bypass alone also rescues the fixed decay (90.4/93.5 at depth five), consistent with error-path attenuation, rather than decay mis-tuning per se, being a principal contributor to the depth collapse; the controller still adds its usual static margin on top of the bypass. We did not re-tune the fixed decay per depth: the claim is that the controller needs no re-tuning, not that no fixed value would serve. A BP+ER reference at the same widths is depth-insensitive ($88.9 \to 87.9$ sequential) but forgets 2× more ($F \approx$ 13–14 vs. 7–10); the local system with skips and the controller beats it at every depth tested. The two starvations are not interchangeable, however: on split CIFAR-100 depth is a net loss that skips only half refund (development-phase numbers: four hidden layers $1.9/1.6 \to 3.8/3.0$ with skips, against 5.7/6.0 at two), whereas *width* at the short chain length lifts the local system to $5.8 \pm 0.9/7.7 \pm 1.1$ (1024–512) — the thin-signal regime wants capacity, not depth, and BP+ER's remaining lead there (10.1 ± 0.4) comes with twice the forgetting.

What the controller must never see, and what it does not do. Composition follows one rule: the drive estimate must be fed only *unamplified waking updates*. Inflated one-hot targets and the offline night's 3×-gain batches both push the EMA up until λ_ℓ hits its clip and erases day learning — in development, night × controller collapsed to 60.2 ± 2.8 against 90.9 ± 3.2 for the night alone — whereas mechanisms that respect the flag compose cleanly (anchor and bout gate under the controller kept the standard signature in development). And the controller deliberately does not subsume λ's second role: on small tabular problems the decay earns its keep as a *capacity* regulariser against overfitting (cf. Appendix B), information that no drive ratio contains; where generalisation, not alignment, sets the correct decay, a tuned constant should keep the job.

8 Discussion

Why micro-batches work. A replay micro-batch is a high-variance gradient estimate. Applied to a static coalition it perturbs the very units encoding the wake stream, and with shared optimiser state the perturbation compounds (the rotation-free unmasked curve of Figure 4). Two things defuse it. Alternating the coalition — the rotation — is where most of the accuracy comes from: with it, even unmasked replay holds 91.5 at batch 16 and 89.3 at batch 4. Confining the update to the mask adds 0.1–0.3 points at those batch sizes, removes the collapse that unmasked replay shows in half the seeds at batch 2 (Appendix H), and gives the guarantee that the present input's hidden computation, under its suppression mask, is untouched — exactly on the proven channels, and for all but $\approx 0.3\%$ of waking samples per step in measurement (Propositions 1–2 and the Remark). The offline night enjoys neither: with nothing clamping the network, a small noisy batch at 3× learning rate moves the whole weight state.

Correspondence with biology. We treat the biology as motivating analogy, not as a claim of mechanistic equivalence. The refractory rule mirrors the causal finding of Driessen et al. (2026) that ON/OFF *alternation*, not tonic reduction, discharges local sleep pressure and rescues memory — our soft-rotation control (a tonic reduction) is reliably worse, and the rotation-free silent mask, with no imposed off-periods at all, is worse still. The dissociation of triggers — homeostatic for waking consolidation, novelty-timed for full nights — is a testable prediction for biology. The regime dependence of the rotation price (cost on easy streams, gain under chronic underfitting) may bear on why local sleep in vivo both impairs behaviour (Vyazovskiy et al., 2011) and serves useful functions when appropriately timed.

Honest limits. Every configuration choice across the project's phases was made on the official test sets — which served as the development set — and every number reported in this paper is measured on a held-out tenth of the training data (training data only during development, never consulted by any selection, never trained on in the reported runs; two independent splits for the headline rows). Development-phase numbers sit 1.1 points above the reported ones on average and rank-correlate with them at Spearman $\rho = 0.97$ (Appendix I). The protocol is five epochs per task, not a single pass (single-pass results in Appendix G). All results are at MLP scale on MNIST/CIFAR variants with three seeds (six on the headline rows); BP+ER — an external reference, not a state-of-the-art claim; stronger replay baselines such as DER++ are not included — keeps a 0.5-point lead on feature-based CIFAR with both sides at their best — a lead the substrate decomposition attributes to dense activation rather than to backpropagation itself; rotation keeps a 1–2-point static price at the record fraction that no gate we tested removes without a sequential cost (Section 6); the same-substrate offline night is highly variable across seeds (89.0 ± 3.5 on the reported split, 91.5 ± 0.5 on the second, where it trails the full system by only 0.3), a phenomenon we report but do not explain; and the novelty claims rest on our own literature search (August 2026); individual primitives — refractory competition, sparse or null-space isolation, replay, adaptive scheduling — all have ancestors, and the claimed contribution is their combination into concurrent, exactly isolated consolidation.

9 Future Work

(i) The bimodal instability of the same-substrate offline night at the record fraction (89.0 ± 3.5) — what makes offline consolidation fragile on wide substrates; (ii) the bout gate's residual static price (94.2 vs. 96.0 rotation-free, development-phase 5% numbers): the OFF-side refractory, utility exemption and synaptic anchoring are now all ruled out as routes past it (Section 6), leaving mastery-annealed rotation depth untried; (iii) the narrow-vs-wide substrate interaction of offline nights at tiny buffers; (iv) convolutional stacks — depth and residual-style routing are answered (Section 7: five hidden layers at the two-layer level with locally learned skips and the controller; on 100-class streams width, not depth, is what pays), convolution and a learned front-end are not, and the static-axis wall of the 100-class regime (6–8% under every substrate change) says the front-end is the binding constraint there; (v) a throughput theory of the isolated channel predicting the required replay cadence from ρ_ℓ, code overlap and the interference rate; (vi) the rotation–spiking calibration interaction, which was optimiser-dependent in development and small on the held-out split; (vii) a drive estimate that survives amplified update streams (inflated targets, night gains), which currently defeat the controller's clip.

10 Conclusion

In the evaluated class-incremental MLP settings, consolidation does not require a separate offline phase. The invariance the isolation provides does not protect a finished forward pass; it closes the channel through which consolidation would perturb the units the present stream is using. Its accuracy value is 2.6 points without rotation and 0.1–0.3 with it (Table 1): once the coalition alternates, the live units tolerate unmasked replay down to four-sample micro-batches, and what isolation adds is the guarantee that they need not — and, at two-sample batches where that tolerance runs out, graceful degradation in place of collapse. With isolated replay, refractory rotation of the active coalition, homeostatic burst triggering and a relative-novelty gate on the rotation itself, a biologically constrained learner consolidates between the steps of its stream: it matches and slightly exceeds the strongest offline-night schedule on split-MNIST at 2.4× the night's replay samples (and still above it at 1.2×), pays one to two points on i.i.d. streams, gains outright on harder data, and runs at a twentieth of dense-rate inference energy when spiking. The offline night retains one irreducible role — consolidating very small episodic stores — which sharpens, rather than blurs, the division of labour between wake and sleep.

References

Abbasi, A., Nooralinejad, P., Braverman, V., Pirsiavash, H., Kolouri, S. (2022). Sparsity and heterogeneous dropout for continual learning in the null space of neural activations. *CoLLAs*, PMLR 199.

Akrout, M., Wilson, C., Humphreys, P., Lillicrap, T., Tweed, D. (2019). Deep learning without weight transport. *NeurIPS 32*.

Arani, E., Sarfraz, F., Zonooz, B. (2022). Learning fast, learning slow: A general continual learning method based on complementary learning system. *ICLR*.

Bazhenov, A., Delanois, J.E., Krishnan, G.P. (2026). Not just after one: Sleep-inspired replay prevents catastrophic forgetting after sequential tasks. *arXiv:2606.08447*.

Borbély, A.A. (1982). A two process model of sleep regulation. *Human Neurobiology* 1, 195–204.

Benna, M.K., Fusi, S. (2016). Computational principles of synaptic memory consolidation. *Nature Neuroscience* 19, 1697–1706.

Bricken, T., Davies, X., Singh, D., Krotov, D., Kreiman, G. (2023). Sparse distributed memory is a continual learner. *ICLR*.

Cayco-Gajic, N.A., Clopath, C., Silver, R.A. (2017). Sparse synaptic connectivity is required for decorrelation and pattern separation in feedforward networks. *Nature Communications* 8, 1116.

Cayco-Gajic, N.A., Silver, R.A. (2019). Re-evaluating circuit mechanisms underlying pattern separation. *Neuron* 101, 584–602.

Driessen, K., Squarcio, F., Tononi, G., Cirelli, C. (2026). Induction of cortical ON/OFF periods in awake mice fulfills sleep functions. *Nature Neuroscience* 29, 1954–1965.

Farajtabar, M., Azizan, N., Mott, A., Li, A. (2020). Orthogonal gradient descent for continual learning. *AISTATS*, PMLR 108.

Golden, R., Delanois, J.E., Sanda, P., Bazhenov, M. (2022). Sleep prevents catastrophic forgetting in spiking neural networks by forming a joint synaptic weight representation. *PLoS Computational Biology* 18, e1010628.

Golkar, S., Kagan, M., Cho, K. (2019). Continual learning via neural pruning. *arXiv:1903.04476*.

Harun, M.Y., Gallardo, J., Hayes, T.L., Kemker, R., Kanan, C. (2023). SIESTA: Efficient online continual learning with sleep. *TMLR*.

Hasselmo, M.E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology* 16, 710–715.

Horowitz, M. (2014). Computing's energy problem (and what we can do about it). *IEEE ISSCC*, 10–14.

Iyer, A., Grewal, K., Velu, A., Souza, L.O., Forest, J., Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurorobotics* 16, 846219.

Kapoor, K., Mackey, W., Aloimonos, Y., Lin, X. (2026). HiCL: Hippocampal-inspired continual learning. *AAAI* 40(27), 22518–22526.

Kaski, S., Kohonen, T. (1994). Winner-take-all networks for physiological models of competitive learning. *Neural Networks* 7, 973–984.

Klasson, M., Kjellström, H., Zhang, C. (2023). Learn the time to learn: Replay scheduling in continual learning. *TMLR*.

Kolen, J.F., Pollack, J.B. (1994). Backpropagation without weight transport. *IEEE ICNN*.

Kubo, Y., Delanois, J.E., Bazhenov, M. (2025). Toward lifelong learning in equilibrium propagation: Sleep-like and awake rehearsal for enhanced stability. *arXiv:2508.14081*.

Lin, E., Luo, W., Jia, W., Yang, S., Xu, W. (2026). Online continual learning via spiking neural networks with sleep enhanced latent replay. In *14th International Conference on Intelligent Control and Information Processing (ICICIP)*, pp. 360–367.

Lässig, F., Aceituno, P.V., Sorbaro, M., Grewe, B.F. (2023). Bio-inspired, task-free continual learning through activity regularization. *Biological Cybernetics* 117, 345–361.

Litwin-Kumar, A., Harris, K.D., Axel, R., Sompolinsky, H., Abbott, L.F. (2017). Optimal degrees of synaptic connectivity. *Neuron* 93, 1153–1164.

Maeda, M., Miyajima, H. (1999). Competitive learning methods with refractory and creative approaches. *IEICE Trans. Fundamentals* E82-A(9), 1825–1833.

Mallya, A., Lazebnik, S. (2018). PackNet: Adding multiple tasks to a single network by iterative pruning. *CVPR*.

Masse, N.Y., Grant, G.D., Freedman, D.J. (2018). Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization. *PNAS* 115, E10467–E10475.

McKee, K., Hazy, T., Zheng, Y., Bugaud, Z., Miconi, T. (2026). Cortex-inspired continual learning: Unsupervised instantiation and recovery of functional task networks. *arXiv:2604.24637*.

Merolla, P.A., et al. (2014). A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science* 345, 668–673.

Payeur, A., Guerguiev, J., Zenke, F., Richards, B.A., Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience* 24, 1010–1019.

Pham, Q., Liu, C., Hoi, S. (2021). DualNet: Continual learning, fast and slow. *NeurIPS 34*.

Robinson, B.S., Lau, C.W., New, A., Nichols, S.M., Johnson, E.C., Wolmetz, M., Coon, W.G. (2022). Continual learning benefits from multiple sleep mechanisms: NREM, REM, and synaptic downscaling. *arXiv:2209.05245*.

Ross, G.J., Adams, N.M., Tasoulis, D.K., Hand, D.J. (2012). Exponentially weighted moving average charts for detecting concept drift. *Pattern Recognition Letters* 33, 191–198.

Rueckauer, B., Lungu, I.-A., Hu, Y., Pfeiffer, M., Liu, S.-C. (2017). Conversion of continuous-valued deep networks to efficient event-driven networks for image classification. *Frontiers in Neuroscience* 11, 682.

Saha, G., Garg, I., Roy, K. (2021). Gradient projection memory for continual learning. *ICLR*.

Saper, C.B., Scammell, T.E., Lu, J. (2005). Hypothalamic regulation of sleep and circadian rhythms. *Nature* 437, 1257–1263.

Shen, Y., Dasgupta, S., Navlakha, S. (2021). Algorithmic insights on continual learning from fruit flies. *arXiv:2107.07617*.

Shen, J., Ni, W., Xu, Q., Tang, H. (2024). Efficient spiking neural networks with sparse selective activation for continual learning. *AAAI* 38, 611–619.

Sorrenti, A., Bellitto, G., Proietto Salanitri, F., Pennisi, M., Palazzo, S., Spampinato, C. (2025). Wake–sleep consolidated learning. *IEEE Transactions on Neural Networks and Learning Systems* 36(7), 12668–12679.

Tadros, T., Krishnan, G.P., Ramyaa, R., Bazhenov, M. (2022). Sleep-like unsupervised replay reduces catastrophic forgetting in artificial neural networks. *Nature Communications* 13, 7742.

Tilley, M.J., Miller, M., Freedman, D.J. (2023). Artificial neuronal ensembles with learned context dependent gating. *ICLR.*

Tononi, G., Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron* 81, 12–34.

Vitter, J.S. (1985). Random sampling with a reservoir. *ACM Transactions on Mathematical Software* 11, 37–57.

Vyazovskiy, V.V., Olcese, U., Hanlon, E.C., Nir, Y., Cirelli, C., Tononi, G. (2011). Local sleep in awake rats. *Nature* 472, 443–447.

Wang, S., Li, X., Sun, J., Xu, Z. (2021). Training networks in null space of feature covariance for continual learning. *CVPR.*

Whittington, J.C.R., Bogacz, R. (2017). An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation* 29, 1229–1262.

A One training step, the three channels and the skip path

Algorithm 1 lists one waking step of the default system with the state each quantity is read from. Two conventions matter for the guarantees. First, the awake sets are read from the waking forward pass, i.e. from the weights before the unmasked waking update, while the masked replay step acts on the updated weights; Propositions 1–2 hold verbatim for a mask computed on the weights being updated, and the drift reported in Section 4.2 is measured under the implemented order. Second, the replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and receive the update (Corollary 1 makes that update invisible to the waking batch for any magnitude), and a burst of n_b micro-batches reuses the mask of its waking batch. The waking step is $\eta_w = \eta\, m/256$ for a waking batch of m samples (the per-epoch drift of a batch-256 protocol), the replay step $\eta_r = 3\eta$ per micro-batch, with $\eta = 0.02$ throughout.

Algorithm 1 One waking step of the default system (rotation always on, one burst per batch).

Require: weights W_t, velocities v_t, suppression $s(t)$, buffer $\mathcal{M}$, pressures S

$a \leftarrow \text{forward}(X_t;\, W_t,\, s(t))$, $\varepsilon \leftarrow \text{sweep}(a, y_t)$ — Eqs. (3)–(4)
$(W, v) \leftarrow$ unmasked heavy-ball step on (W_t, v_t) with $G^\ell = \varepsilon^\ell (a^{\ell-1})^\top$, rate η_w — wake update
$\mathcal{A}^\ell \leftarrow \{i : \exists b,\ a^\ell_{i,b} \neq 0\}$ for every hidden layer; $S_i \leftarrow S_i + \bar{a}_i(X_t)$ — from the pre-update a
$M^\ell \leftarrow$ Eq. (6), unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell]$, readout row and bias unmasked
$s^\ell_i(t{+}1) \leftarrow \mathbf{1}[i \in \mathcal{A}^\ell]$ — refractory rotation
write (X_t, y_t) into $\mathcal{M}$ (reservoir)
for each of the n_b replay micro-batches $(X_r, y_r) \sim \mathcal{M}$ **do**
  $a_r \leftarrow \text{forward}(X_r;\, W,\, s{=}0)$, $\varepsilon_r \leftarrow \text{sweep}(a_r, y_r)$ — suppression removed
  $(W, v) \leftarrow$ masked step, Eq. (7), with $G^\ell = \varepsilon^\ell_r (a^{\ell-1}_r)^\top$, rate η_r, masks M^ℓ
end for
$S_i \leftarrow 0$ for every $i \notin \mathcal{A}^\ell$ if a burst ran — discharge
return $W_{t+1} = W$, $v_{t+1} = v$, $s(t{+}1)$

The three channels. Table 3 lists the three ways a synapse enters the mask (6) and what each guarantees; the rotation of Section 4.3 exists to move synapses from the second row into the third.

Channel	Condition	Invisible to X	Source
presynaptic silent	$j \notin \mathcal{A}^{\ell-1}(X)$	for any magnitude	Prop. 1
postsynaptic naturally asleep, active pre	$i \notin \mathcal{A}^\ell(X)$, $s^\ell_i = 0$	under the margin $\tau^\ell_{k,b}$	Prop. 2
postsynaptic suppressed	$s^\ell_i = 1$	for any magnitude	Cor. 1

Table 3: The three consolidation channels opened by the synapse mask.

A numerical example. On one sample with $k = 2$ and $\sigma(z) = (3, 2, 0.5, 0)$, units 3 and 4 are asleep and $\tau_k = 2$. Any replay update with $\sigma(z_3 + \delta_3) < 2$ and $\sigma(z_4 + \delta_4) < 2$ leaves the activity vector $(3, 2, 0, 0)$ unchanged, because the two winners' own inputs are not touched; if unit 4 is refractory ($s_4 = 1$) its bound is void and δ_4 may be arbitrarily large; and every synapse leaving unit 4 into the next layer may move by any amount, since its presynaptic activity is zero.

Skip synapses. For $\ell \geq 3$ each deep hidden layer receives a bypass, $z^\ell = W^\ell a^{\ell-1} + S^\ell a^{\ell-2} + b^\ell$, with S^ℓ under the same Dale projection and a feedback twin $B^\ell_S \in \mathbb{R}^{n_{\ell+2} \times n_\ell}$ that carries the bypass

Table 4: Substrate ablation on i.i.d. MNIST (256–128 hidden core, three seeds; development phase, official test set). "Syn. ops" is the fraction of the dense MLP's synaptic operations in one forward pass. Upper block: constraints added cumulatively; lower block: one ingredient replaced at its stage.

Substrate	Accuracy	Syn. ops
dense backpropagation MLP (autograd reference)	97.7 ± 0.1	1.00
local burst-error learner, unconstrained	97.8 ± 0.1	0.85
+ bounded error, Dale's law, sign-concordant feedback (no transport)	97.1 ± 0.0	0.85
+ k-WTA 10%	97.0 ± 0.2	0.80
+ relaxation T=5	97.0 ± 0.1	0.80
+ 30% distance-dependent connectivity	96.5 ± 0.2	0.24
+ burst gate and burst baseline	96.3 ± 0.2	0.24
+ single-phase burst sweep replacing relaxation (*final substrate*)	96.5 ± 0.1	0.24
random wiring instead of distance-dependent, equal density	95.4 ± 0.0	0.24
one settling step (T=1) instead of five	17.9 ± 6.8	0.80
weight transport restored (mirroring, no KP decay)	96.3 ± 0.2	0.24
Kolen–Pollack decay removed	96.3 ± 0.1	0.24
heavy-ball SGD instead of Adam ($\eta = 0.02$)	95.2 ± 0.3	0.24
k-WTA 5% / 2% instead of 10%	95.9 ± 0.1 / 94.2 ± 0.2	0.79/0.78
multiplicative burst coding	78.8 ± 1.9	0.24
training on T_s=8 spike counts	80.0 ± 0.8	0.24

target's burst back to its source,

$$\varepsilon^\ell \leftarrow \varepsilon^\ell + \sigma'(z^\ell) \odot \Big((B_S^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+2}/\kappa) \odot \mathbf{1}(a^{\ell+2} > 0) \big] \Big),$$

S^ℓ and $B_S^{\ell-2}$ learned by the local product of (5) with the target layer's decay λ_ℓ, and replay-masked by the same rule two layers apart, $M_{S,ij}^\ell = \mathbf{1}[\, j \notin \mathcal{A}^{\ell-2}(X) \ \vee \ i \notin \mathcal{A}^\ell(X)\,]$, so that Propositions 1–2 extend unchanged.

B The price of the substrate

Table 4 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST where each constraint's cost is cleanest. The upper block is cumulative; the lower block replaces one ingredient at its stage and is to be compared with the matching cumulative row.

The cumulative ladder prices the entire biological constraint set at 1.2 points below the dense backpropagation reference (97.7 → 96.5) for 4× fewer synaptic operations per pass; the local burst-error objective itself costs nothing (97.8 unconstrained). No single constraint is expensive: the bounded-error/Dale/no-transport trio costs 0.7, 10% k-WTA one tenth of a point, and the 70% connectivity cut half a point — of which distance-dependent wiring recovers +1.1 over random wiring at equal density. The lower block adds three cautions. Settling cannot merely be shortened (T=1 collapses to 17.9); the single-phase burst sweep is the principled replacement, matching the T=5 accuracy in a single pass. The Kolen–Pollack decay and even fully restored weight transport are accuracy-neutral at this scale — the decay earns its keep as a regulariser on harder inputs, not as an aligner. And the two spiking-flavoured training schemes fail outright: spiking is affordable at *inference* (the energy measurements above), not as a training-time code.

Finally, the two tables disagree about the optimiser in an instructive way: on the plain substrate SGD *loses* 1.3 points to Adam (95.2 vs. 96.5), yet inside the isolated-consolidation loop it wins on

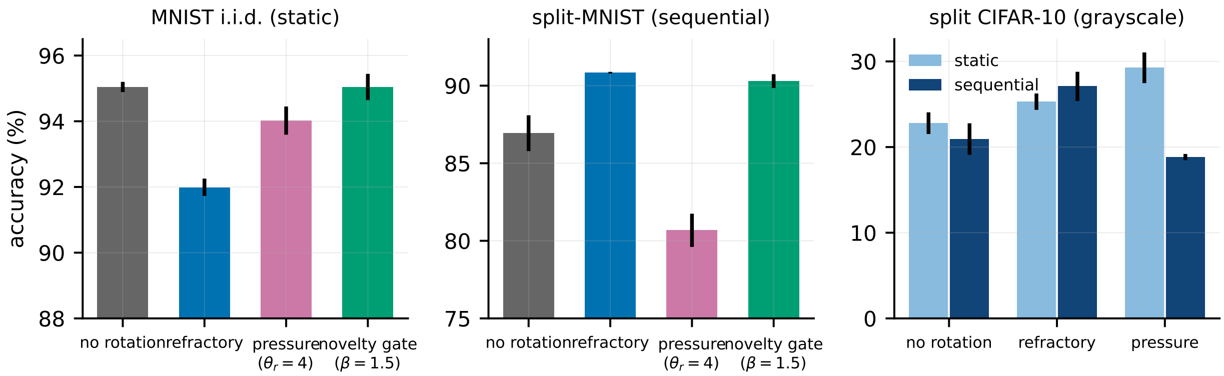

Figure 7: Rotation: price, trade-off, and gate. Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 (grayscale) rotation is a gain on *both* axes.

both axes. Per-synapse adaptive state is an asset while every synapse trains on every batch and a liability once updates become intermittently mask-confined — the staleness mechanism of Table 1's middle block seen from the other side. The optimiser is not a neutral implementation detail: it interacts with the sleep machinery, and the right choice reverses between the static substrate and the consolidating system.

C The per-batch rotation gates under masked Adam

Appendices C–G report development-phase numbers (official test set) from the earlier 5% phases, the small-buffer and down-selection studies and the learning-rate sweeps; they were not re-run under the held-out protocol and support mechanism orderings only.

Figure 7: always-on rotation costs 1.3–3.0 points on i.i.d. MNIST (95.0 $\to$ 92.0–93.7). Gating rotation by *unit-level pressure* only trades the axes (static 91.6 $\to$ 94.0 as sequential 88.8 $\to$ 80.7; no dominant point). The *relative novelty* gate (8) removes the price at no sequential cost on MNIST: static $\mathbf{95.0 \pm 0.4}$ (rotation on 1.4–17% of batches), sequential 90.3 ± 0.4 (β=1.5) or 90.9 ± 0.6 (β=1.1). On split CIFAR-10 the price *reverses*: rotation gains +2.5 (refractory) to +6.5 points (pressure-selected rest) on the static axis — forced rotation acts as a regulariser in the underfit regime — while the pure novelty gate (8) *misfires* there (sequential 19.1 ± 1.1: fast/slow $\to 1$ on a stream that never becomes predictable, so rotation shuts off exactly where it helps). The progress-keyed gate (9) resolves this with one setting across all four regimes: split CIFAR-10 sequential 27.2 ± 0.7 and static 25.3 ± 1.0 (both = always-on rotation), split-MNIST 90.4 ± 0.9, i.i.d. MNIST 94.3 ± 0.2 at γ=0.5 — retaining a residual 0.7-point static cost relative to the pure novelty gate on easy streams, the price of the unmastered clause. (All gate numbers in this subsection use masked Adam; under heavy-ball SGD the gate must be committed to minimum-duration bouts — Section 6.)

D Small buffers: what the night still buys

At K=200 the night appeared to keep a $\sim$2.6-point edge in our earlier phases; the final triangulation (Figure 8) dissolves most of it into a substrate mismatch. On the *same* 512–256/5% substrate, the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest

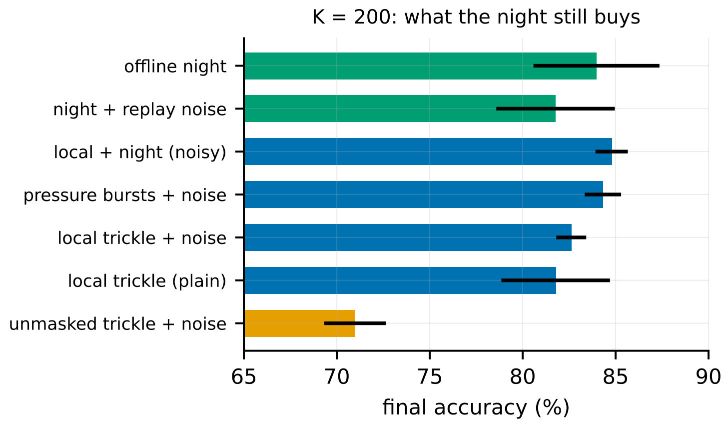

Figure 8: $K = 200$ **mechanism triangulation.** Isolated rotating replay beats the unmasked trickle, diversity helps only during wake, and the irreducible advantage of the night at tiny buffers is its interference-free window.

tiny-buffer number remains the narrower 256–128/10% night (86.9), so the offline window retains an advantage only in combination with narrow substrates. The mechanism results stand: replay diversity (noise σ=0.2) is specifically a wake-side lever — it lifts every local scheme ($81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$) and *hurts* the night ($84.0 \rightarrow 81.8 \pm 3.2$) — and *unmasked*, rotation-free waking replay collapses (71.0 ± 1.7): isolated rotating replay beats it at every buffer size we tested.

E Sleep-anchored down-selection

Motivated by the synaptic homeostasis hypothesis (Tononi and Cirelli, 2014), we anchored structural plasticity (prune the weakest fraction of live synapses, regrow the same number with a distance bias) to consolidation windows, using the Kolen–Pollack decay of (5) as the natural weakening force on unreplayed synapses. The verdict splits three ways at matched turnover ($\approx$ 5% per epoch). On the *continuous* local-sleep substrate pruning is free (sleep-anchored 90.9 ± 0.2, clocked 90.6 ± 0.4, none 90.8 ± 0.1). On the pure *night* substrate, pruning immediately after each night helps substantially (87.2 ± 3.8 vs. 84.1 ± 1.3) — offline down-selection rescues the wide-sparse night. On the *episodic-burst* substrate pruning is harmful (84.6 ± 0.5 burst-anchored, 86.6 ± 1.3 clocked, vs. 89.9 ± 0.1 without): when replay is rare, synapses are cut before enough consolidation events have protected them. Down-selection, like replay itself, needs either density or an offline window.

F Learning-rate sensitivity

The optimiser comparison invites an objection: SGD's η was chosen, Adam's was left at its default. A two-sided sweep dissolves it. At the 10% record fraction (three seeds), heavy-ball SGD under always-on rotation holds $92.2 \pm 0.7/93.3 \pm 0.2$ at η=0.01, $92.7 \pm 0.3/94.7 \pm 0.3$ at 0.02 (the joint optimum) and $91.5 \pm 0.6/94.7 \pm 0.3$ at 0.05, decaying at 0.005 (91.3/91.3) and destabilising at 0.1 (79.5 ± 17.3), while masked Adam has its optimum at the default ($91.8 \pm 0.6/94.6 \pm 0.2$ at 10^{-3}; 89.9/89.9 at $3{\cdot}10^{-4}$, 90.9/94.3 at $3{\cdot}10^{-3}$): the comparison of Table 1 is at each optimiser's best rate. The original sweep, at the 5% fraction of the earlier phases, told the same story. There, under always-on rotation, SGD holds a sequential plateau over $\eta \in [0.01, 0.02]$ ($91.9 \pm 0.5/92.0 \pm 0.6$) and a static plateau over $[0.02, 0.05]$ (93.3–93.7), decaying gracefully outside (86.3/89.2 at η=0.1);

η=0.02 is the joint optimum. Masked Adam, swept over a tenfold range, has its sequential optimum exactly at the default (90.8 ± 0.1 at 10^{-3}; 88.6 at $3{\cdot}10^{-4}$, 88.9 at $3{\cdot}10^{-3}$) — the comparison of Section 5.3 was already at Adam's best — and no Adam setting reaches the heavy-ball joint point ($3{\cdot}10^{-3}$ buys 93.6 static at a three-point sequential cost). Unmasked SGD is poor and unstable at *every* rate (84–88, seed deviations 1.8–4.1, three- to eight-fold those of the masked runs), making the stabilising role of isolation plus rotation (the unmasked runs are rotation-free) a cross-η fact. The bout-committed gate inherits the plateau, and the optimum shifts one notch down on CIFAR (η=0.01).

G Probes of the residual static price

Single-pass streams. With one epoch per task instead of five (everything else as in the headline cells, three seeds), refractory local sleep reaches 91.8 ± 0.2 (F=5.7) against 91.6 with five epochs, BP+ER 88.7 ± 0.3, the offline night 75.1 ± 3.5 (one night per task is too few), unmasked interleaved replay 76.0 ± 20.1 and no buffer 18.1 (held-out split): the proposed system keeps its lead over every tested control in a single pass, while the offline night falls below BP+ER.

Bout length and the OFF-side refractory. Bout length is a proper knob: M=512 (duties 0.61/0.38) gives $91.7 \pm 0.4/94.0 \pm 0.5$, already recovering most of what the per-batch gate loses, while M=4096 drives the static duty to 0.94 and the refund vanishes (93.4 ± 0.1). The static ceiling of silent isolation (96.0) stays out of reach — rotation, even in bouts, keeps a residual price — so the committed gate buys regime-adaptivity, not a free lunch. Completing the flip-flop with an OFF-side refractory (once a bout expires, triggers are ignored for M_{off} batches) is a clean null: short refractories ($M_{\text{off}} \leq 1024$) never engage on the static stream — a bout expires only after its trigger has been quiet for M batches, so static re-triggering is not flicker to begin with — while a long one (M_{off}=4096) lowers the static duty to 0.53 without moving static accuracy (94.0 ± 0.7) and monotonically erodes the sequential axis ($92.1 \rightarrow 91.0$, occasionally masking a task switch). Together with the M=512 cell (duty 0.38, static 94.0), this locates the residual static price in the rotation of settled coalitions itself, not in how often the gate fires.

Structural exemption and synaptic anchoring. *Utility-exempt rotation* spares the top-q units by long-term use trace from ever being benched, so the settled coalitions stay awake. It traces a smooth trade-off between always-on rotation and the rotation-free system (q=0.10: $90.9 \pm 0.1/94.2 \pm 0.2$; q=0.25: $89.1 \pm 0.4/95.4 \pm 0.2$), but the bout gate beats the q=0.10 point by +1.1 sequential at equal static: *temporal commitment dominates structural exemption*, recapitulating at the utility level what unit-level pressure gating showed earlier (Figure 7) — any structural narrowing of the rotation pool is paid for in replay-channel width.

Two-timescale synapses give every weight a slow anchor in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016): the fast weight decays toward its anchor at rate λ while the anchor absorbs it at rate μ, ticking only on waking updates so the exactness of the replay path is untouched. The strong hypothesis — that settled function living in the anchor would dissolve rotation's static price — is falsified: across $\lambda \in [3{\cdot}10^{-4}, 3{\cdot}10^{-3}]$, $\mu \in [10^{-4}, 10^{-3}]$ no cell exceeds the anchor-free static accuracy, and strong coupling hurts both axes (the anchor becomes a lagging drag). Weak symmetric coupling (λ=μ=$3{\cdot}10^{-4}$), however, acts as a *sequential stabiliser*: 92.4 ± 0.2, the best sequential cell of the 5% grid with the seed variance halved, at static parity — weakly dominating the 5% always-on point. Anchor and bout gate compose sub-additively ($92.3 \pm 0.3/93.8 \pm 0.5$, between the parents).

H Isolation across replay batch sizes

Figure 9 extends the left panel of Figure 4 to two-sample replay micro-batches and shows, per batch size, the seed-paired margin of the full system over rotation with unmasked replay (held-out split; six seeds at batches 2, 4 and 16, three at 8). From 16 down to 4 the margin is 0.1 $[-0.3, +0.5]$, 0.3 $[-0.2, +0.6]$ and 0.3 $[-0.5, +1.2]$: isolation costs nothing and buys nothing measurable in mean accuracy while the rotation keeps the live coalition tolerant of unmasked replay. At batch 2 the replay gradient is too noisy for either system — the isolated one falls to 79.0 ± 3.2 (forgetting 21.5 ± 4.2 against 6.4 at batch 16), every seed between 75.5 and 84.3 — but the unmasked one bifurcates: three seeds finish at 82.5–84.8, slightly *above* their isolated counterparts (+0.3, +1.8, +8.1), one at 58.2 and two at chance (9.9). Leaking replay onto the live coalition is therefore not a pure loss — the surviving seeds show that the extra plasticity can help — but it opens a failure mode that the mask closes: with the current input's hidden computation held invariant, the noisiest replay degrades the system gracefully instead of destroying it. The paired mean at batch 2 (+24 $[-0.2, +48.5]$) describes that failure mode, not a margin, and is kept out of the ranges quoted in the main text.

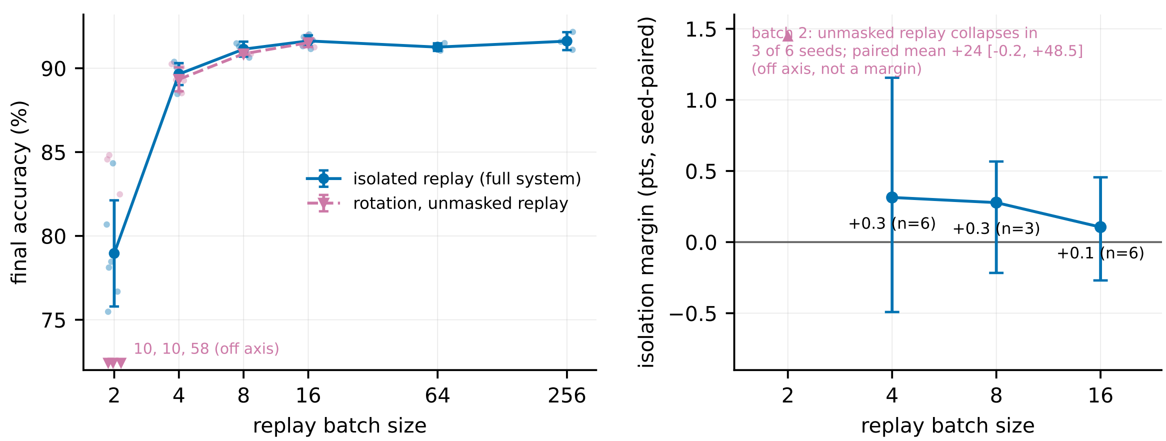

Figure 9: Isolation across replay batch sizes (held-out split). **Left:** split-MNIST accuracy against replay batch size for isolated replay (the full system) and for rotation with unmasked replay, every seed shown; at batch 2 no unmasked mean is drawn because three of six seeds fail (values at the axis floor). **Right:** the seed-paired margin of isolation with a 95% bootstrap interval at the batch sizes where the unmasked run does not collapse.

I Development-phase numbers and a second held-out split

Every configuration choice made during this project — some four hundred sequential-axis configurations over twenty phases — was made on the official test sets, which served as the development set; the paper reports numbers measured on a stratified tenth of the training data held out from training and never used for any decision, with the remaining nine tenths as the stream. Table 5 places the development-phase numbers beside the reported ones and beside a second, independent held-out tenth (a different split seed) for the headline rows. Over the 87 sequential configurations run under both protocols, reported numbers sit 1.1 points below the development numbers on average (a smaller training set and a distribution-matched evaluation set both contribute) and rank-correlate with them at Spearman $\rho = 0.97$; every ordering carried by the paper holds on both held-out splits, with one magnitude that does not: the same-substrate night, bimodal on the first split (89.0 ± 3.5), reaches 91.5 ± 0.5 on the second and trails the full system by 0.3 there. The adaptive decay's static margin on MNIST is likewise split-dependent (93.5 ± 4.3 with one collapsed seed on the first split, 95.2 ± 0.4 on the second). Table 6 gives the per-task accuracy matrix of the headline configuration on the reported split.

Table 5: Development-phase (official test set) numbers beside the reported held-out numbers and a second held-out split. Sequential accuracy, with the static axis on its own row where the paper reports it; six seeds on the reported split for the headline rows, three elsewhere. Lower block: raw split CIFAR-10.

	development (test)	reported (held-out)	second split
always-on rotation (default)	92.7 ± 0.3	91.6 ± 0.3	91.8 ± 0.4
static	94.7 ± 0.3	94.1 ± 0.2	94.1 ± 0.6
bout-committed gate	92.3 ± 0.2	91.8 ± 0.3	92.0 ± 0.6
rotation alone (unmasked replay)	92.1 ± 0.5	91.5 ± 0.3	91.8 ± 0.6
static	94.7 ± 0.1	93.7 ± 0.3	93.7 ± 0.3
silent mask (no rotation)	88.8 ± 0.8	88.0 ± 0.7	88.6 ± 1.3
unmasked interleaved ER (SGD)	87.0 ± 1.1	85.4 ± 3.5	80.0 ± 14.1
BP + ER	89.3 ± 0.8	88.8 ± 0.3	88.5 ± 0.7
offline night, same substrate	90.9 ± 3.2	89.0 ± 3.5	91.5 ± 0.5
offline night, narrow substrate	90.7 ± 0.4	89.2 ± 1.7	89.3 ± 0.6
adaptive λ (drive)	91.7 ± 0.2	90.6 ± 0.6	90.8 ± 0.1
static	96.1 ± 0.2	93.5 ± 4.3	95.2 ± 0.4
five hidden layers + skips + adaptive λ	91.5 ± 0.4	91.0 ± 0.3	91.1 ± 0.3
static	96.2 ± 0.3	95.8 ± 0.3	95.7 ± 0.1
no buffer	19.6 ± 0.0	19.4 ± 0.0	19.5 ± 0.0
split CIFAR-10 (raw)			
rotation	29.5 ± 0.8	28.9 ± 1.0	29.7 ± 0.9
offline night	26.2 ± 0.6	25.7 ± 1.2	26.5 ± 0.4
BP + ER	25.1 ± 1.5	24.8 ± 0.3	25.8 ± 0.8
unmasked ER	17.0 ± 6.1	15.4 ± 4.2	15.5 ± 4.5
no buffer	16.4 ± 0.2	16.3 ± 0.1	16.6 ± 0.0

Table 6: Per-task accuracy matrix of the headline configuration on split-MNIST (held-out split, mean over six seeds): row = after training task i, column = accuracy on task j. Forgetting F, the mean of the diagonal minus the last row over tasks 1–4, is 6.4 ± 0.6; the drop is concentrated on tasks 2–3 and the first task is retained.

after task	T1	T2	T3	T4	T5
1	99.8				
2	98.7	97.2			
3	97.7	92.7	97.0		
4	97.5	89.6	92.9	97.5	
5	96.7	88.1	89.7	91.5	91.8